Internet of Things and Vehicular Networks

TP-546

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What is a city?

- A city is a human settlement located within a defined geographic area.
- It contains many closely spaced buildings used for housing and for cultural, commercial, industrial, financial, and other activities not directly related to land exploitation.

A city is where we live

Cities organize everyday life around homes, neighborhoods, and shared services.

A city is where we work

Cities concentrate jobs, businesses, institutions, and productive activity.

A city is where we connect

Cities provide places where people meet, celebrate, and build community.

A city is where we enjoy life

Culture, leisure, sport, and entertainment are essential parts of urban life.

Citizens are the starting point for change

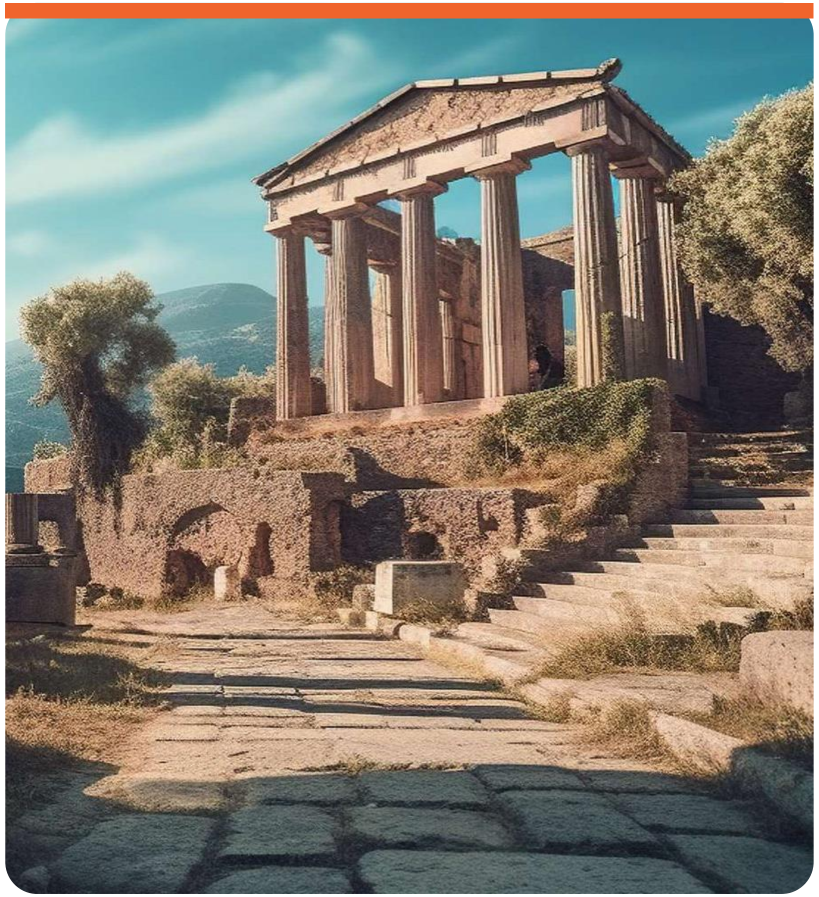
In everyday life, citizens imagine ways to improve their cities. The citizen should therefore be the precursor of urban change.

Urban change begins with participation

People experience city problems directly. Their ideas, reports, and participation help define which improvements matter most.

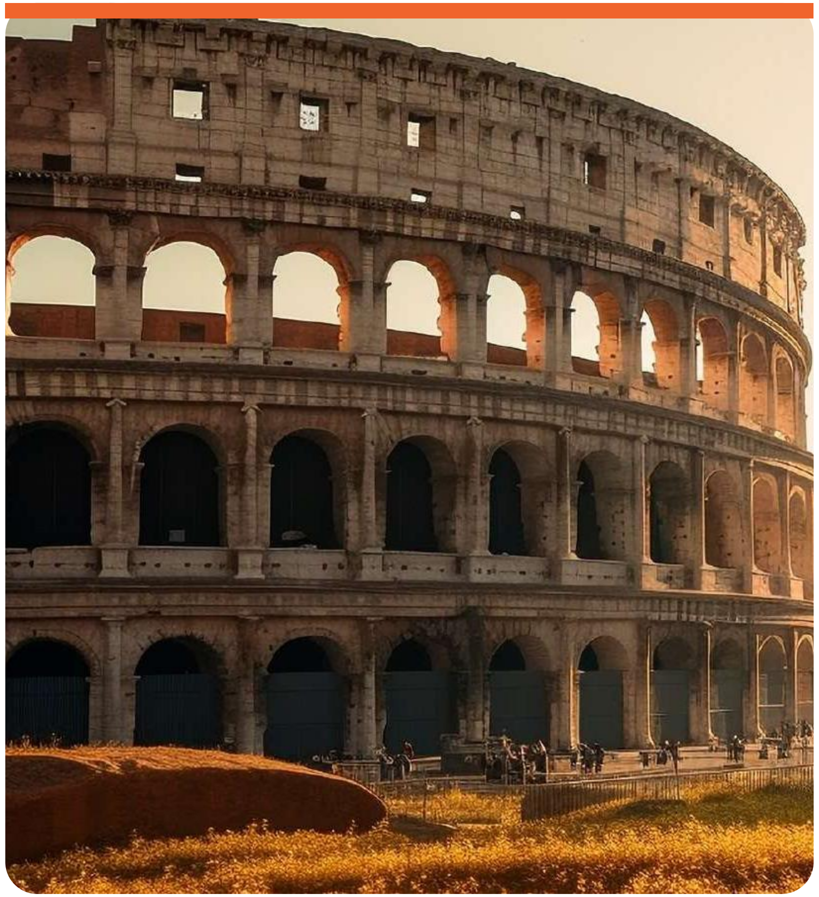
Greek cities

Early Greek cities connected public life, governance, commerce, culture, and shared urban spaces.



Roman cities

Roman cities expanded organized streets, public buildings, water systems, and infrastructure across large territories.

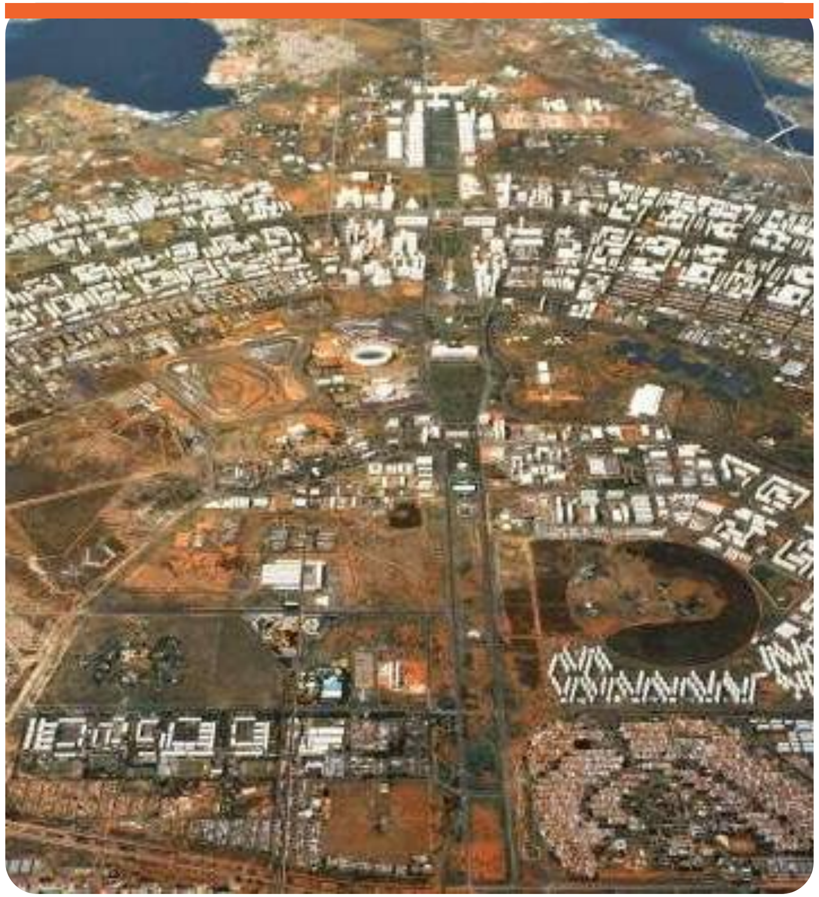


The Industrial Revolution

Industrialization accelerated urban growth and introduced new challenges involving density, mobility, pollution, and public health.

Planned cities

Urban planning coordinates land use, infrastructure, transport, housing, services, and long-term growth.



People belong at the center

According to the Inter-American Development Bank (IDB), a smart city places people at the center of development and incorporates information and communication technologies into urban management.

Technology should strengthen public value

- Digital technologies can support efficient government, collaborative planning, and citizen participation.
- Smart cities promote integrated and sustainable development, becoming more innovative, competitive, attractive, and resilient while improving people's lives.

First generation: technology-led

- Technology vendors encouraged cities to adopt solutions for security, traffic monitoring, and related services.
- The implications of those technologies - including their effect on quality of life - were rarely examined.
- Example: IBM.

Second generation: city-led

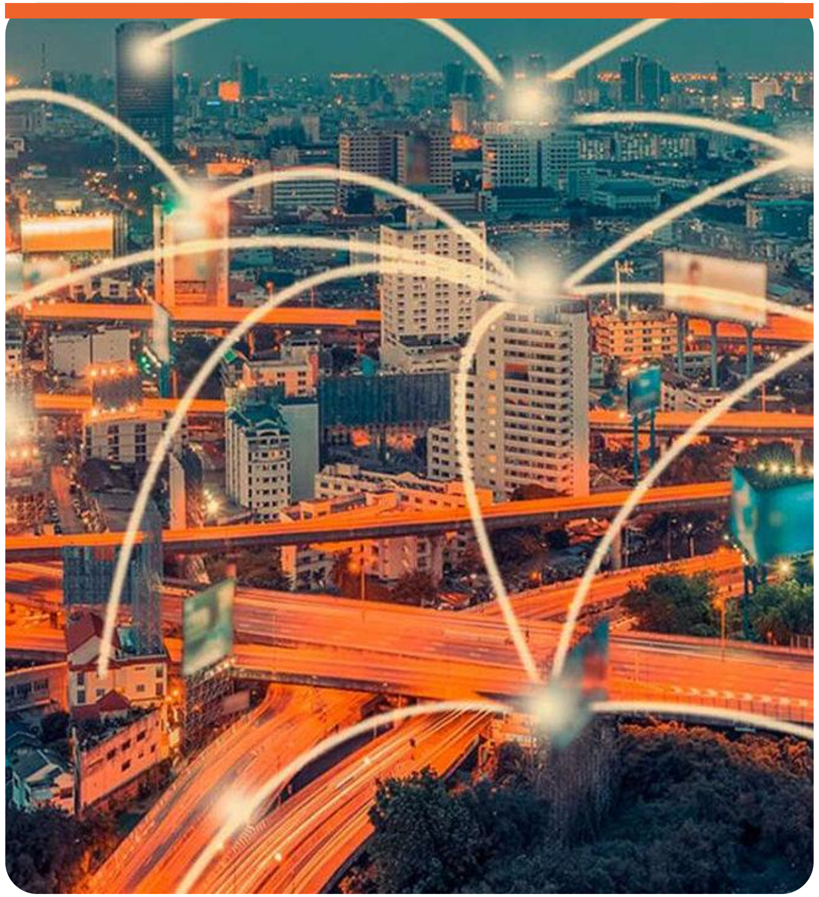
- Municipal government takes the lead instead of technology companies.
- The city defines its desired future and the role of smart technologies and other innovations.
- The focus is on enabling better quality of life for citizens.

Third generation: co-created

- Cities adopt co-creation models with citizens to shape the next generation of smart urban development.
- Equity and social inclusion become central design principles.

Connected data should improve outcomes

Smart cities use technology and connected data to:
- Improve the efficiency of city services.
- Improve quality of life for everyone.
- Increase equity and prosperity for residents and businesses.



A citizen-centered creation cycle

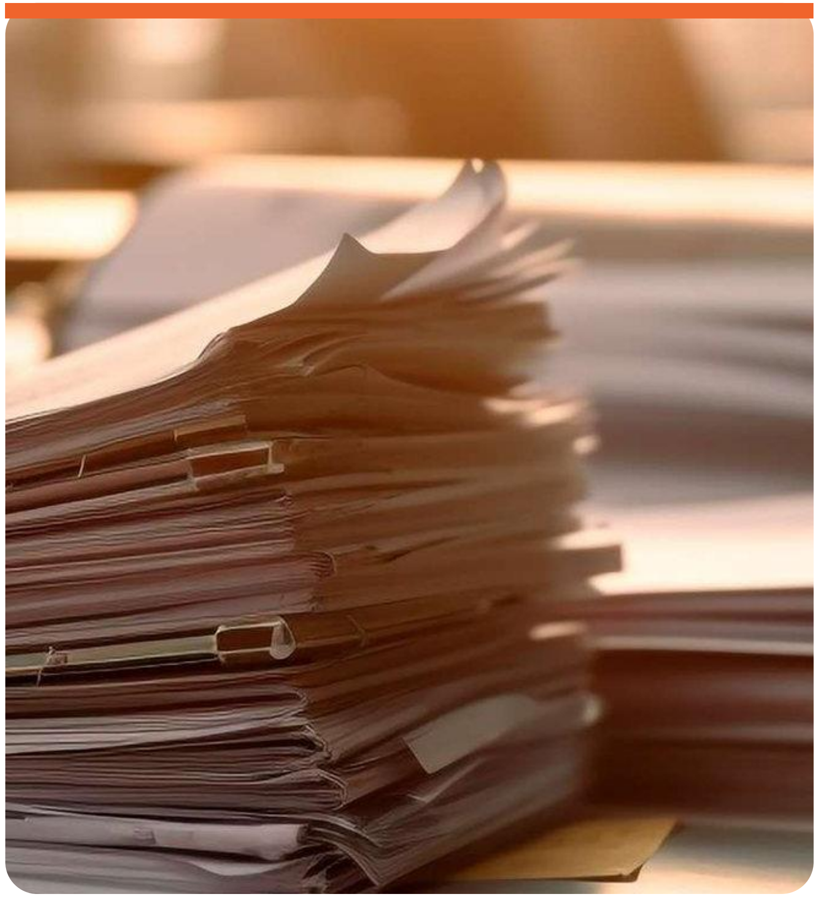
- Meet with the community: give citizens tools and encourage participation.
- Identify and solve: municipal authorities must respond to reported problems.
- Build livable cities: effective public action increases citizen satisfaction.

Measure, prioritize, and act

- City indicators: increase transparency by sharing key performance indicators.
- Rank and decide: prioritize the most important issues.
- Create the next smart-city solutions: take the necessary action without delay.

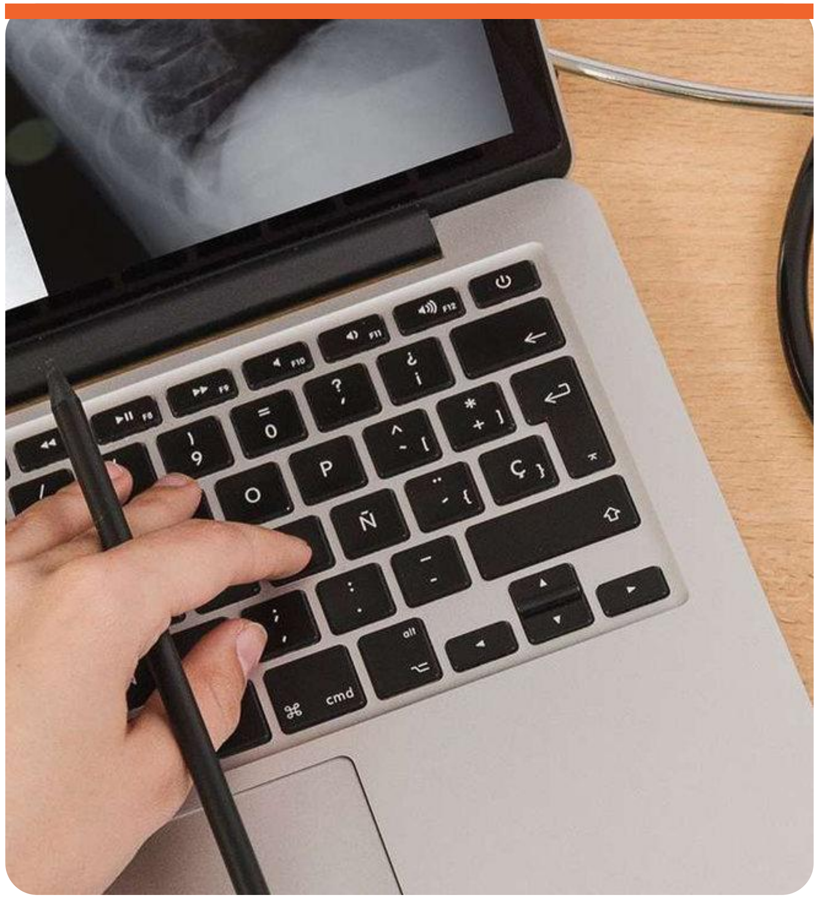
Governance and public administration

- Make government services easier to obtain, including requests, permits, and licenses.
- Increase public transparency.



Health

- Make it easier to schedule medical appointments and examinations.
- Use electronic health records to improve continuity and access to care.



Economic development

A smart city should promote economic development through local plans for industry, innovation, and entrepreneurship, improving productivity and reducing the time needed to start a business.

Simpler processes create opportunity

The easier it is to develop new projects and companies, the more practical and efficient the city becomes - while also creating new employment opportunities.

Environment

Key priorities include:
- Carbon dioxide and methane emissions.
- Waste treatment.
- Access to safe drinking water.

Education

- Improve teaching through different technologies.
- Create virtual learning environments.

Connected learning

- Apps allow parents to monitor student performance.
- Game-based learning platforms can increase engagement.

Urban planning

- Establish guidelines for property construction.
- Define where different activities may operate within the city.

What is a planned city?

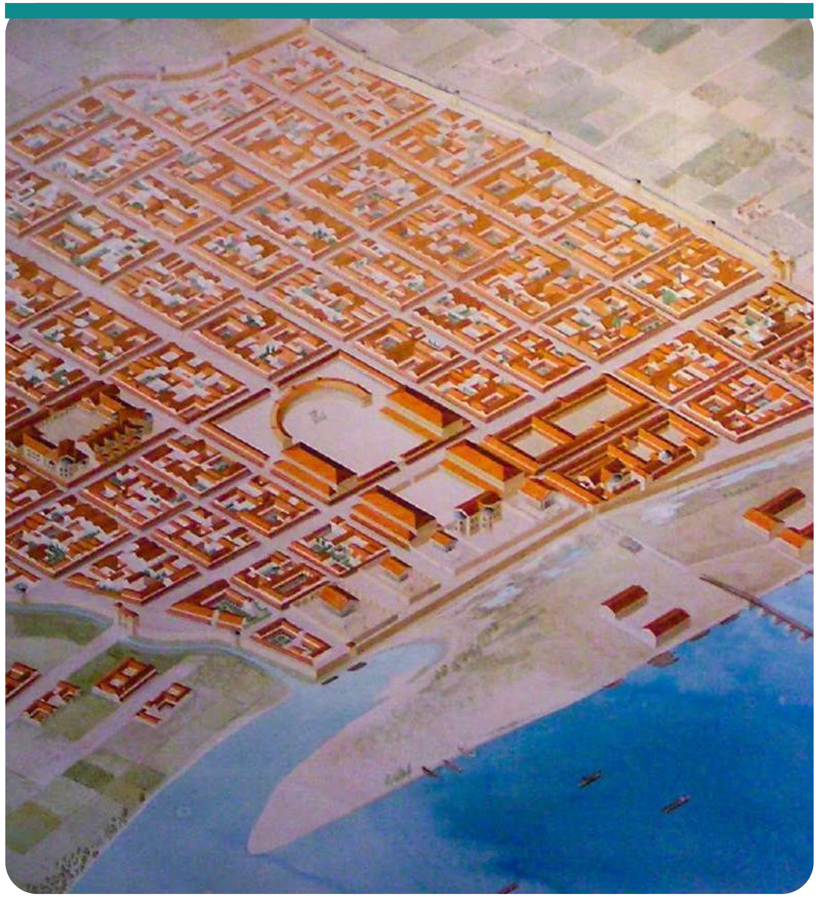
A planned city is built according to a predefined project, usually prepared by a multidisciplinary group of specialists such as architects, urban planners, and engineers.

Planning serves explicit goals

Planned cities may be designed to:
- Improve residents' quality of life.
- Reduce environmental impact.
- Promote economic development.

Medieval urban patterns

During the Middle Ages, many European cities were built using grid patterns, central squares, and radial streets.



Renaissance planning

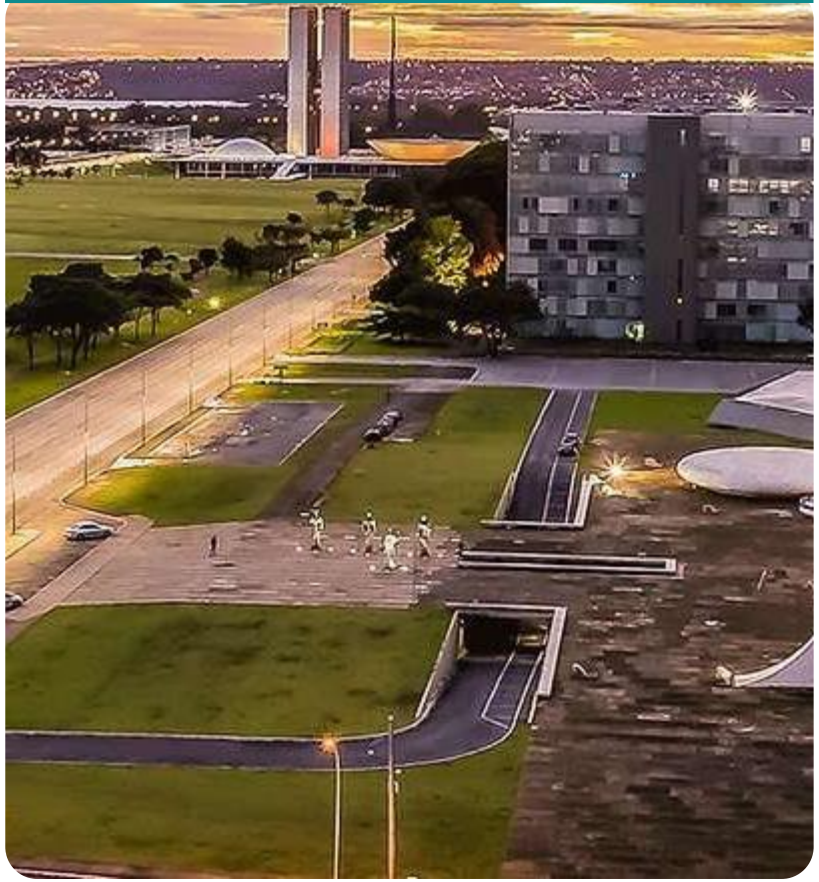
- The Renaissance renewed interest in urban planning.
- Venice, for example, developed a canal system that facilitated the movement of goods and people across the city.

Industrial growth exposed urban limits

- The Industrial Revolution rapidly increased the population of European cities.
- Overcrowding, pollution, and poor living conditions became widespread.
- Urban planners responded by designing cities intended to be healthier and more efficient.

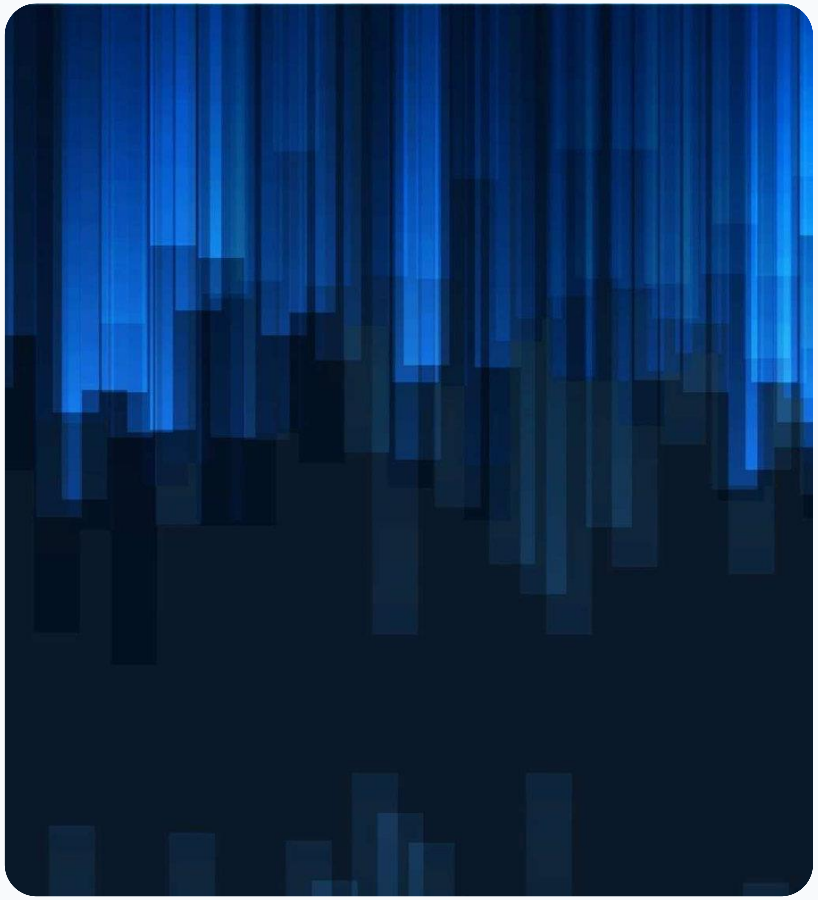
Planning expanded worldwide

During the 20th and 21st centuries, many planned cities were created around the world.



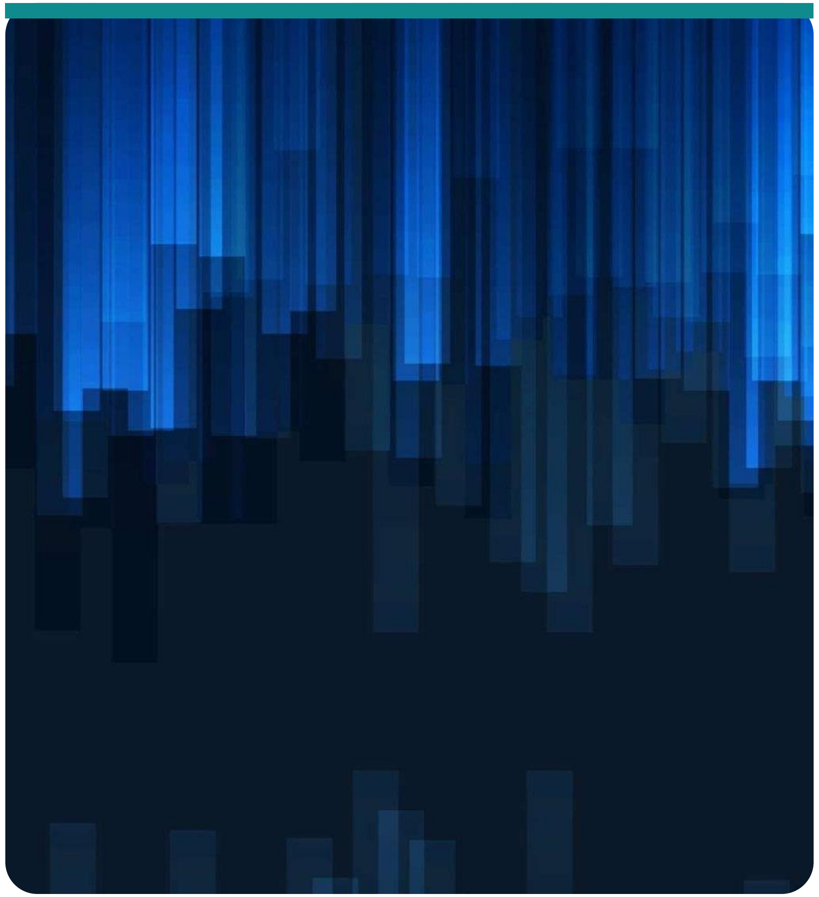
Planned cities face financial and social challenges

- Cost: building a planned city from scratch is expensive and requires funding from multiple sources.
- Public support: residents must be willing to move to a new city and embrace its distinctive features.



Construction requires complex coordination

Planning and coordinating a new city demands significant expertise and alignment across infrastructure, design, finance, regulation, public services, and community needs.



Partnership, sustainability, and participation

- Public-private partnerships can provide funding and expertise while building support.
- Sustainable design can reduce costs and environmental impact.
- Community engagement helps ensure that the city meets local needs.

Step 1: define the city's purpose

Before designing the city, clarify:
- Its goals and objectives.
- Its intended uses.
- The population it is designed to accommodate.
- Its economic-development objectives.

Step 2: select the location

5. Select an area for the city.
6. Evaluate whether the location supports the city's intended uses, population, infrastructure, and future economic growth.

Step 3: develop the master plan

7. Develop a city master plan.
8. Define the overall layout and development strategy.
9. Include land-use zoning, transportation planning, and infrastructure planning.

Urban planning transforms places

Urban planning develops solutions to improve or regenerate an existing urban area, or to create a new urban development in a defined region.

Poor planning creates systemic risk

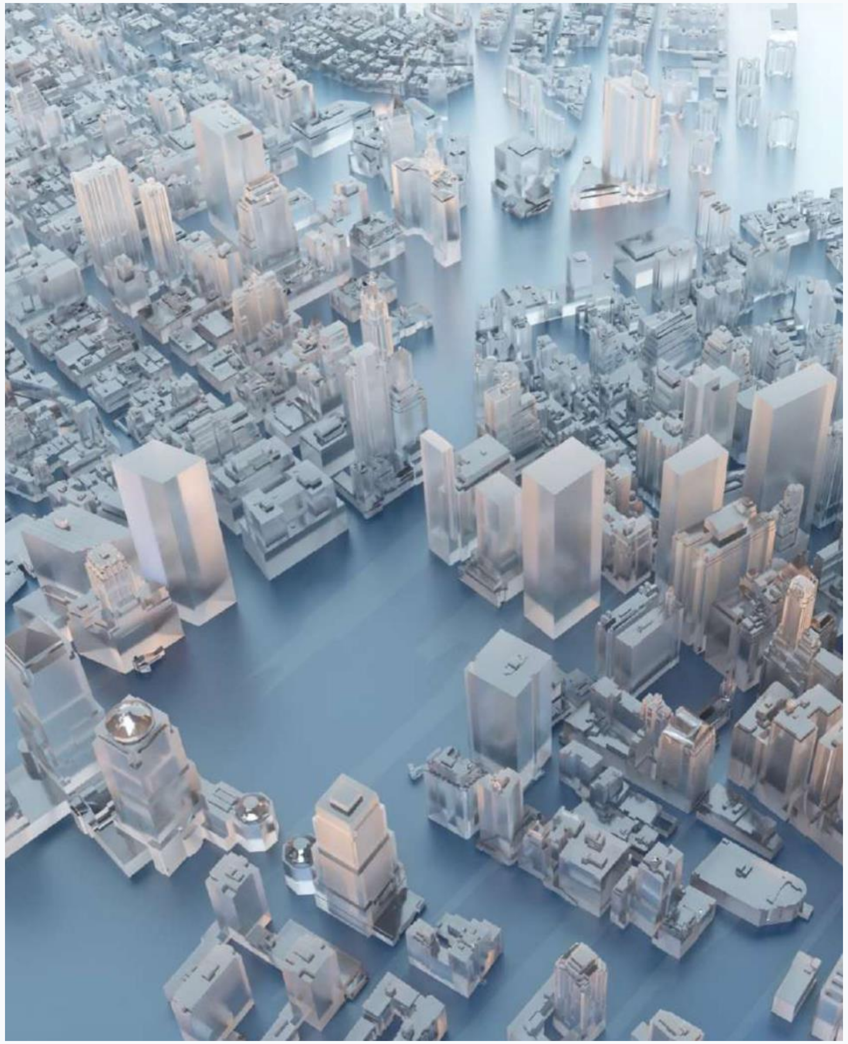
A lack of urban planning can cause traffic congestion, inadequate infrastructure, and greater vulnerability to climate change - including flooding.

Sustainability includes essential services

Building a sustainable planned city involves more than infrastructure. The city must also provide access to high-quality education and healthcare.

A master plan guides long-term growth

- A master plan is essential for the city's operation, future generations, and municipal expansion.
- It can guide both planned cities and existing cities.
- In Brazil, cities with more than 20,000 residents must have a master plan under the City Statute and Article 182 of the Federal Constitution.



The master plan connects territorial policies

The master plan is the basic instrument of municipal territorial-development policy. It integrates planning actions for:

- Leisure.
- Housing.
- Environmental sanitation.
- High-quality urban infrastructure.

Public services belong in the master plan

The master plan also coordinates actions for:
- Transportation.
- Education.
- Health and social assistance.
- Public services for present and future generations.

Prioritize sustainable multimodal mobility

- Develop multimodal transportation systems that support walking, cycling, and public transit.
- Design streets and public spaces that prioritize pedestrians and cyclists.

Design resilient infrastructure

- Build infrastructure capable of withstanding natural disasters.
- Apply resilient design to buildings and essential systems.
- Develop emergency-response plans and systems that protect residents.

Manage water for a sustainable future

- Conserve water through rainwater harvesting, greywater recycling, and low-flow fixtures.
- Use efficient irrigation for landscaping and green spaces.
- Treat and reuse industrial and municipal wastewater.

Turn waste management into a circular system

- Reduce waste through recycling and composting.
- Improve collection and disposal systems.
- Adopt waste-to-energy solutions such as incineration and biogas generation.

Salvador

- Designed by Portuguese architect Luis Dias.
- A geometric grid plan resembling a fortress.

Goiania

- Often described as Brazil's greenest capital.
- Part of Getulio Vargas's plan to build infrastructure in the Central-West region.
- Associated with the 'March to the West' policy.

Brasilia

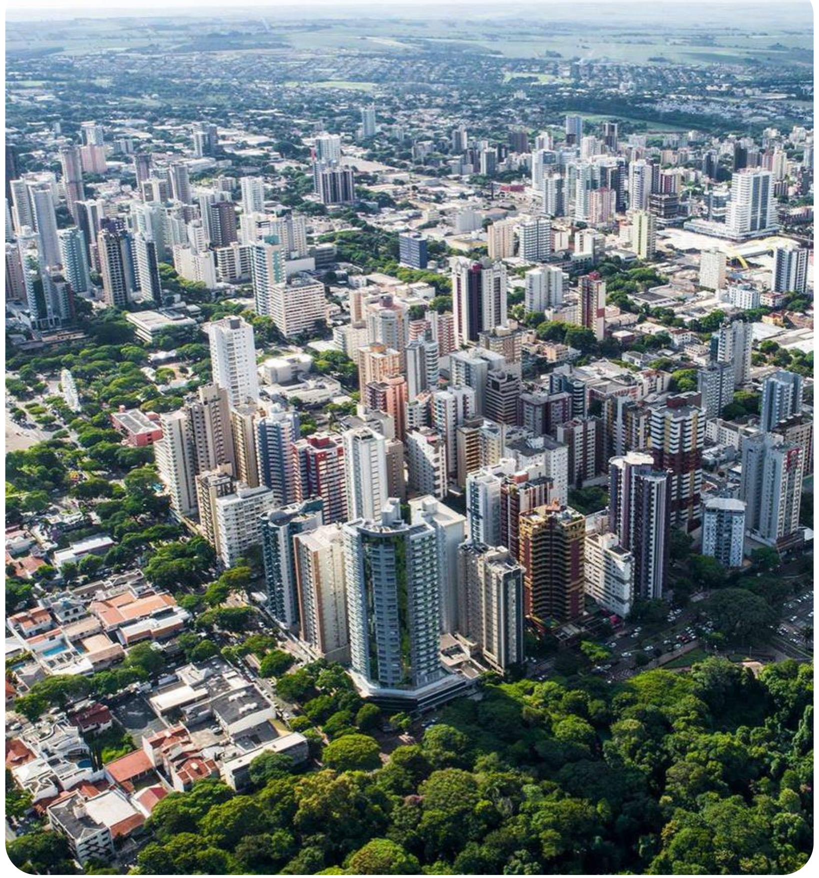
- Brazil's federal capital, designed by urban planner Lucio Costa and architect Oscar Niemeyer.
- Organized around superblocks.
- Clearly defined zoning areas.

Boa Vista

- Designed by civil engineer Alexio Derenusson.
- Inspired by the urban layout of Paris.
- Streets radiate in a fan-shaped pattern.

Maringa

- The 1943 plan was created by Sao Paulo urban planner Jorge de Macedo Vieira.
- Designed as a garden city.
- Features clearly defined zoning.

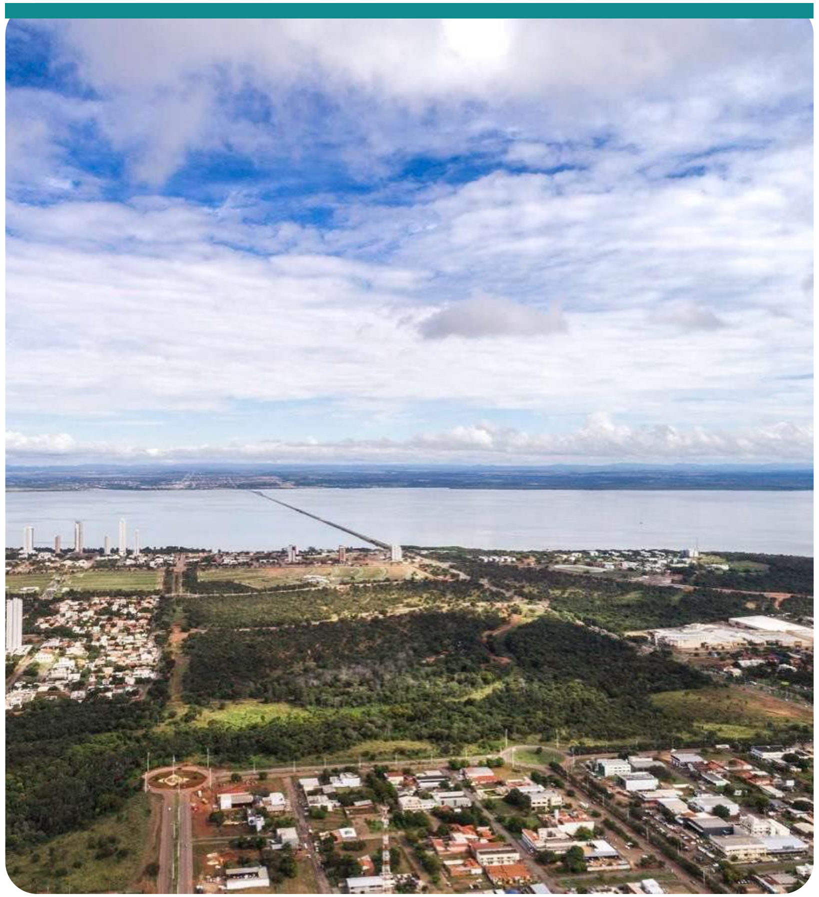


Belo Horizonte

- Planned by engineer and urban planner Aarão Reis.
- Inspired by Paris.
- Characterized by broad avenues.

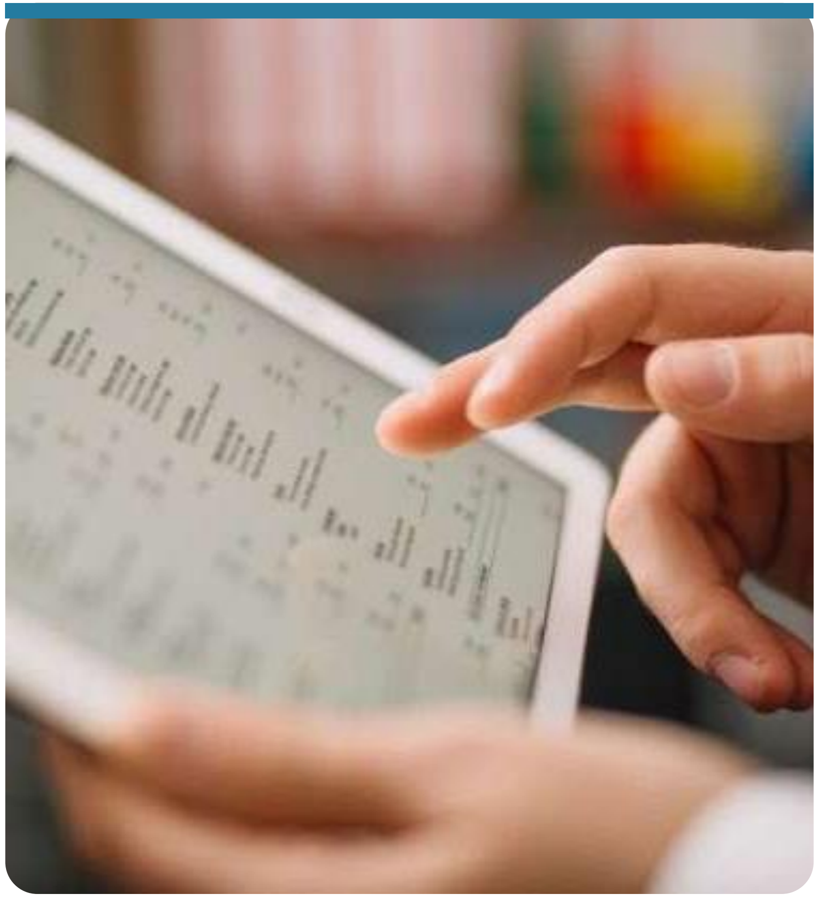
Palmas

- Created as the capital of the new state of Tocantins in 1988.
- Designed by Luiz Fernando Cruvinel Teixeira and Walfredo Antunes de Oliveira Filho, with wide streets and a grid layout.
- Planned for one million people; the source material reports a current population of about 300,000.



Technology must fit the local context

- Technology and infrastructure: adapt solutions to local needs while ensuring compatibility and interoperability.
- Data and governance: build public trust through privacy protection and ethical data use.



Sustainability must be equitable

- Balance environmental concerns with social needs and ensure equitable access to benefits.
- Engage citizens in decision-making and build partnerships for successful implementation.

Small cities can become smarter

- Improve connections within the city.
- Encourage information sharing.
- Enable real-time response.
- Advance sustainability initiatives.
- Promote safety and economic growth.

Small cities need inclusive services

Small cities particularly need social-service solutions that improve quality of life and provide access to education and healthcare comparable to that available in large cities.

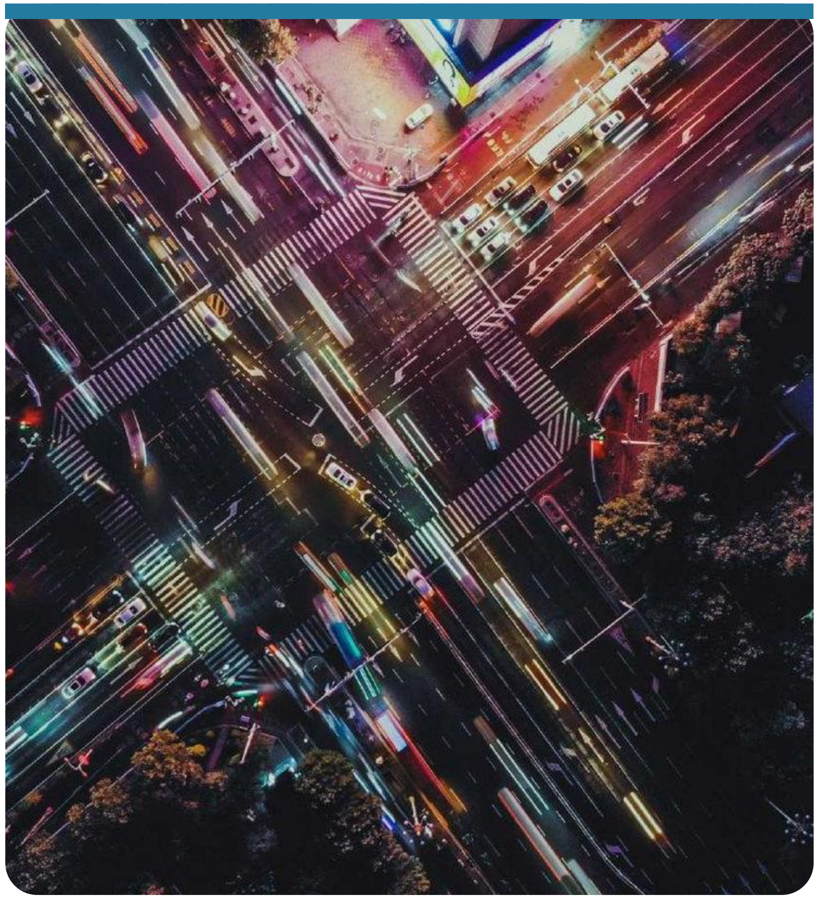
Small and rural cities

Challenges: limited resources, outdated infrastructure, and unique environmental conditions.

Solutions: low-cost sensor networks, precision agriculture, smart water management, and community-led initiatives.

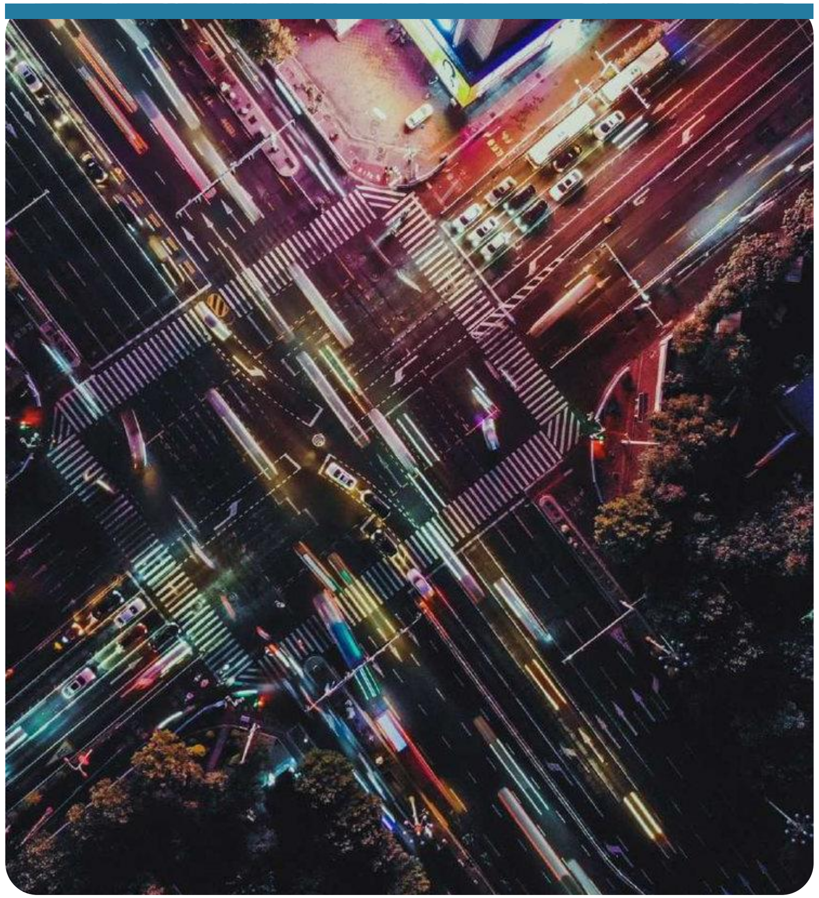
Songdo, South Korea: technology by design

- Built from scratch on reclaimed land near Incheon as a living experiment in sustainability and technology integration.
- Building sensors monitor energy use and adjust settings automatically.
- Electric buses and bicycles provide clean transportation, while smart grids prioritize renewable energy.



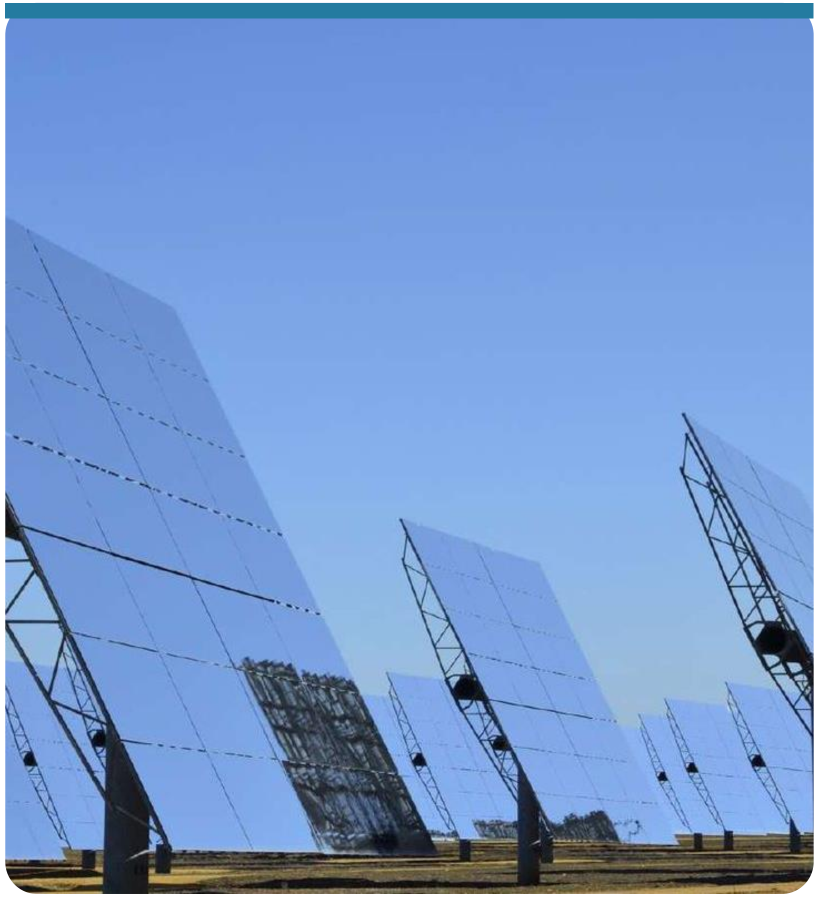
Songdo embeds learning into the city

- Interactive displays and digital games in public spaces create informal learning opportunities for all ages.
- Schools use immersive technologies, including virtual reality, for field trips and simulations.



Hobart, Australia: a smarter energy grid

Tasmania's capital implemented a smart grid that integrates renewable sources such as solar and wind. It reduces dependence on fossil fuels and enables residents to generate and sell clean energy.



Angers, France: digital public engagement

- A mobile app lets residents report problems, access city services, and receive personalized notifications.
- Sensors monitor noise pollution and air quality, enabling targeted interventions.

Angers turns the city into a learning space

- Students use augmented reality to explore historic sites and local landmarks.
- The city app also connects users with educational resources, museums, and libraries, supporting lifelong learning.

Medium-sized cities focus on service efficiency

Solutions typically improve public-service infrastructure, increase the energy efficiency of lighting, streamline waste management, and optimize public-vehicle operations.

Copenhagen, Denmark

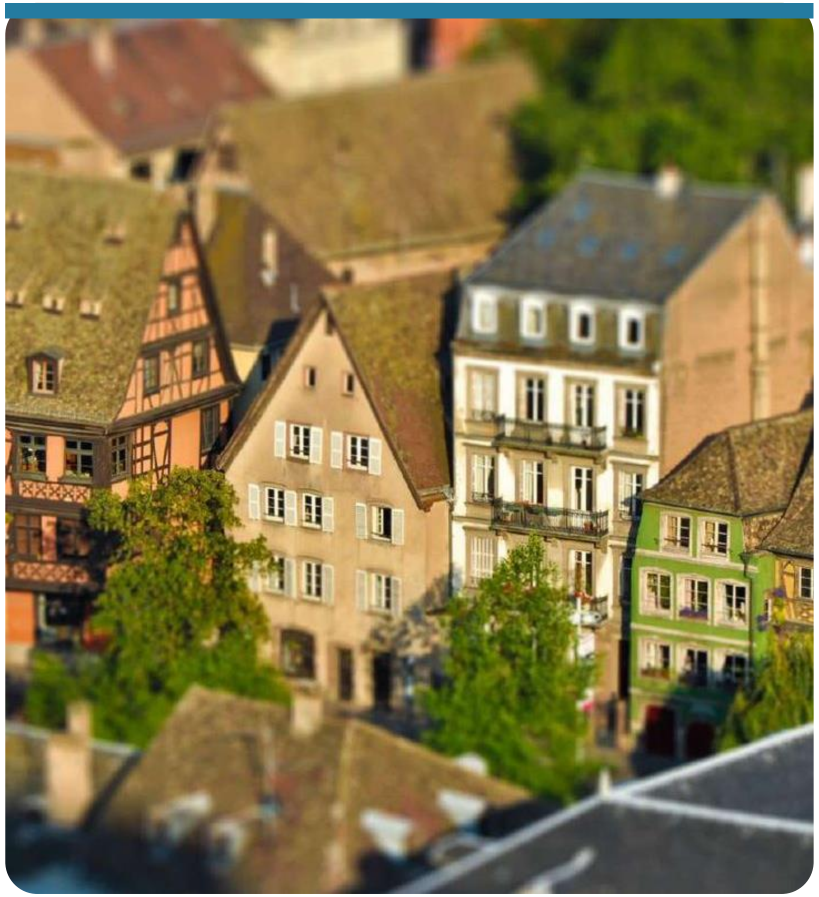
- More than half of residents commute by bicycle.
- Smart traffic lights and bike lanes adapt to real-time demand.
- Electric vehicles and car-sharing programs offer sustainable alternatives.
- Energy-efficient buildings support ambitious carbon-neutrality goals.

Amsterdam, the Netherlands

- Sensors monitor canal levels and enable dynamic flood-prevention measures.
- Smart grids and electric vehicles support the goal of carbon neutrality by 2050.
- World-class cycling infrastructure and pedestrian-friendly streets prioritize public well-being.

Utrecht, the Netherlands

- Utrecht Smart City focuses on mobility, energy efficiency, and citizen engagement.
- Sensors monitor parking availability and air quality.
- Charging stations support electric vehicles.
- Data optimizes waste collection and public-transport routes.



Bologna, Italy

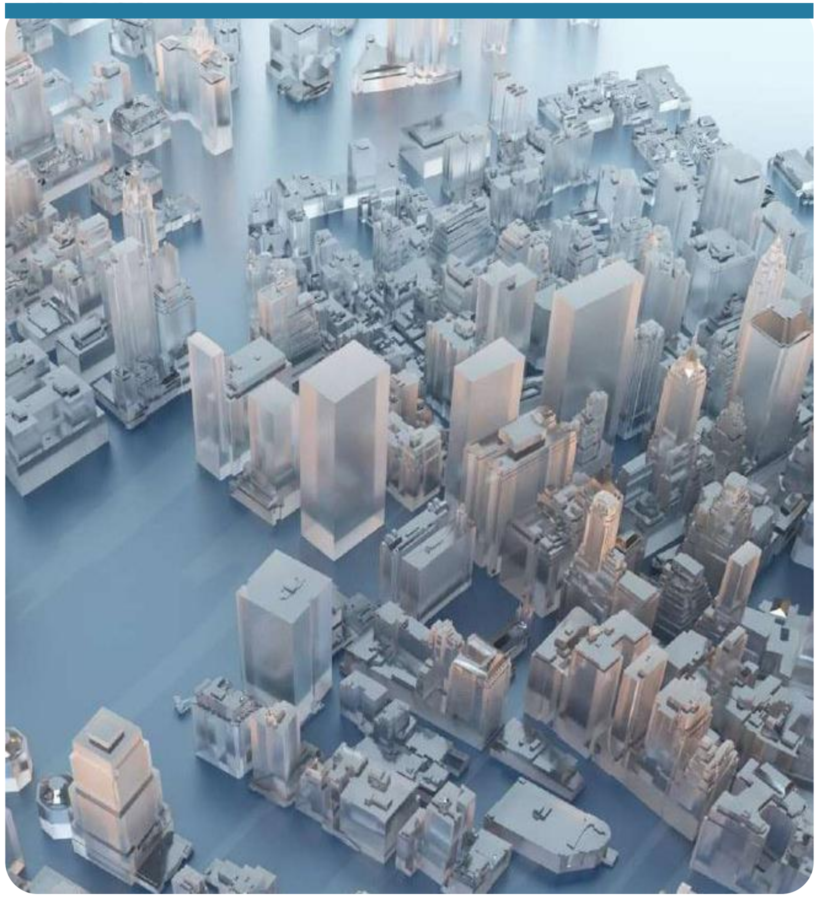
- Sensors monitor the structural health of historic buildings and air quality in heritage areas.
- Telemedicine services improve access to healthcare.
- Smart parking solutions help reduce congestion.

Large cities face challenges of scale

Population and spatial scale increase pressure on transportation and make urban-infrastructure management more complex.

Large-city priorities

The most important solutions improve transportation control, energy consumption, and the effectiveness of public-safety measures.



Singapore

- Frequently ranked among the world's leading smart cities.
- Uses intelligent traffic-management systems to coordinate mobility.

Singapore connects sustainability and digital government

- Green buildings reduce energy consumption.
- Sensor networks monitor air quality and waste levels.
- Digital services and citizen-engagement platforms strengthen governance.

New York City, United States

The 'Big Apple' addresses congestion and sustainability directly through systems that provide real-time traffic and transit information.

New York also prioritizes energy and inclusion

- The city promotes green buildings and renewable energy.
- Smart grids optimize energy distribution.
- Digital-inclusion initiatives broaden access to urban services.

London, United Kingdom

- Prioritizes public transport and environmental sustainability.
- Smart meters and energy-efficiency programs seek to reduce carbon emissions.
- Air-quality sensors and green spaces improve the urban environment.

Seoul, South Korea: data-driven governance

Seoul is a leader in smart technology and data-based governance. Its Data City Seoul platform provides open data for public use, promoting transparency and citizen engagement.

Seoul uses sensors to inform policy

- Sensors track conditions ranging from air quality to noise levels.
- The data supports policy decisions and resource allocation.
- Smart parking and electric-vehicle charging networks complement the system.

Barcelona at a glance

- Capital of Catalonia, Spain.
- Population in the source material: 4.84 million.
- Internationally known for art and architecture.

Barcelona

Urban transformation prepared Barcelona to combine quality of life, global visibility, public space, and digital innovation.

Passeig Maritim

- A three-kilometer waterfront was created for tourism and recreation.
- Bars, restaurants, retail, and public spaces activate the shoreline.

El Poblenou

- Revitalization of a former industrial district once called the 'Catalan Manchester.'
- Development of the Olympic Village.
- New residential, commercial, and leisure areas.

Ring roads reorganized mobility

- New ring roads helped relieve city traffic.
- Key corridors include Ronda de Dalt and Ronda Litoral.

Municipal parks expanded public space

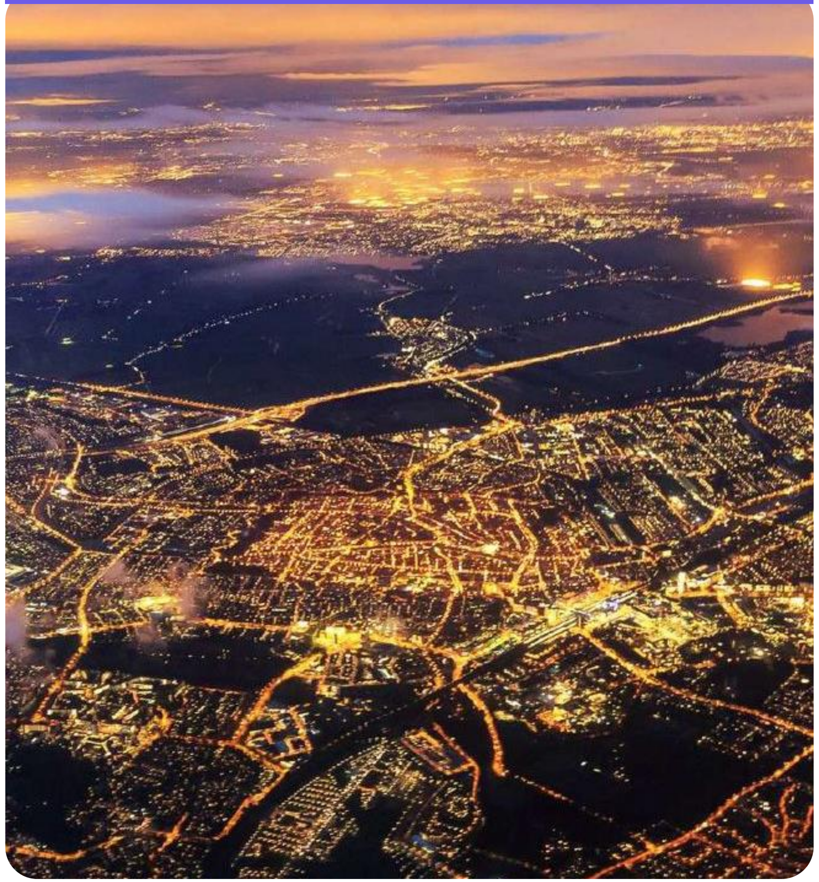
Barcelona invested in municipal parks as part of its broader urban-regeneration strategy.

Technology entered municipal management

- Joan Clos served as mayor from 1997 to 2006.
- Technology was used to improve city management.
- The Ciutat Vella district was revitalized.
- Barcelona hosted the 2004 Universal Forum of Cultures.

Barcelona Smart City 2.0

- The city promoted a new urban-services economy.
- Its vision combined self-sufficient, productive neighborhoods at human speed with a hyperconnected, zero-emission metropolitan area.
- Barcelona hosts the Smart City Expo World Congress.

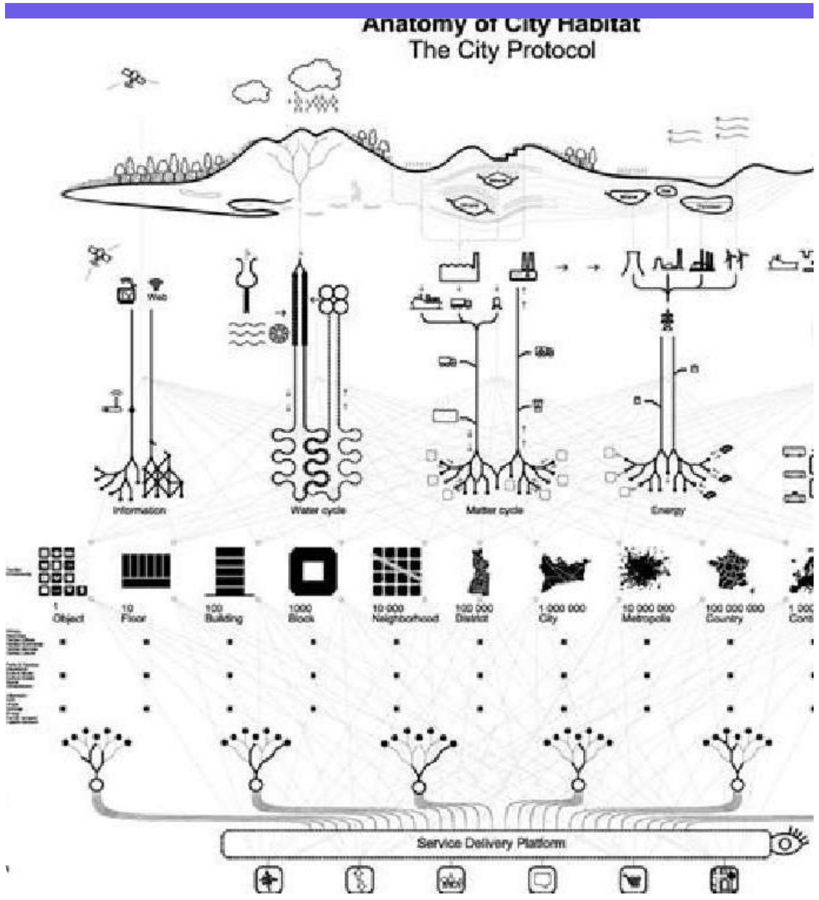


The 22@ innovation district

- Transformed 200 hectares of former industrial land in Poblenou.
- Concentrates incubators, research centers, universities, and companies.

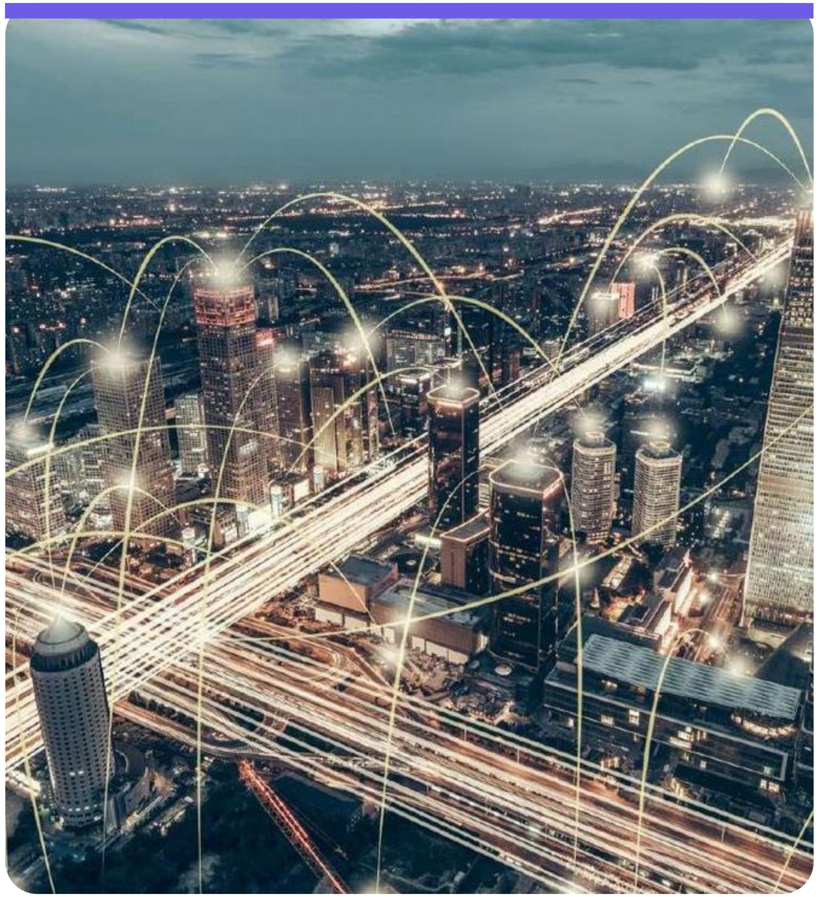
Anatomy of the city

The Barcelona Protocol organizes the city as a system of people, places, infrastructure, information, and urban services.



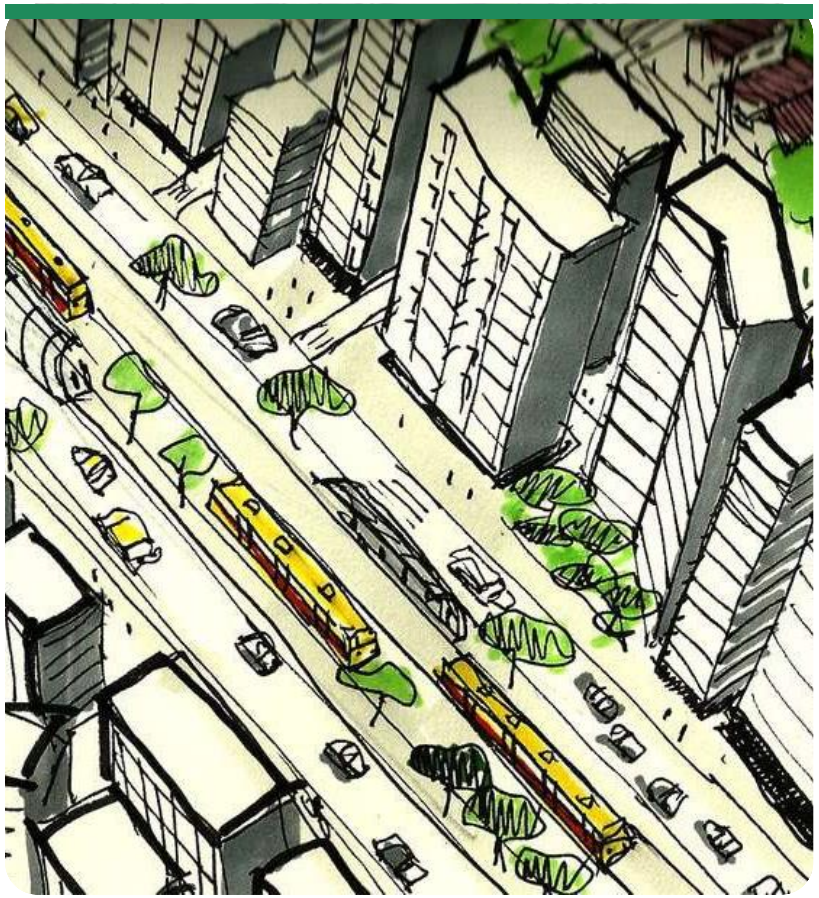
Barcelona Smart City 3.0

- Include local communities in development.
- Use technology to improve quality of life.
- Place data and citizens at the center.
- Expand technical knowledge and digital inclusion.
- Use forums and online applications to support important decisions.



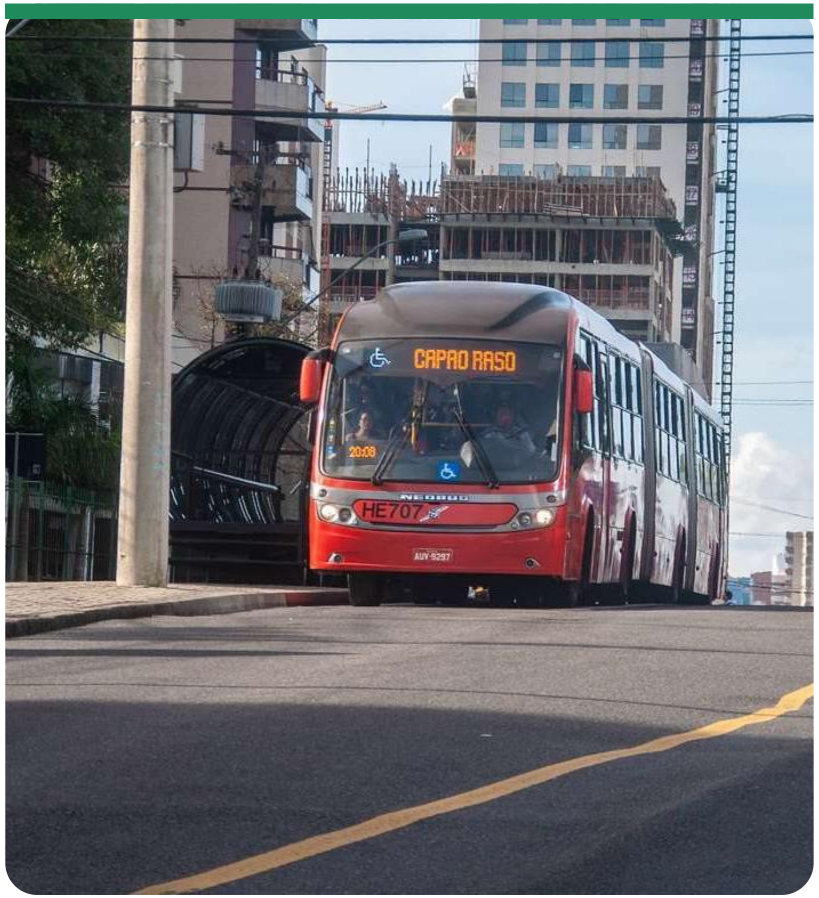
Curitiba: urban planning as a foundation

Curitiba's smart-city trajectory is strongly connected to long-term urban planning.



Bus corridors and tube stations

Public transportation uses exclusive bus corridors and tube stations designed for efficient boarding and disembarking.



Public parks

The city created public parks that combine recreation, environmental protection, and urban resilience.

Waste recycling

Curitiba developed recycling programs that connect environmental management with public participation.

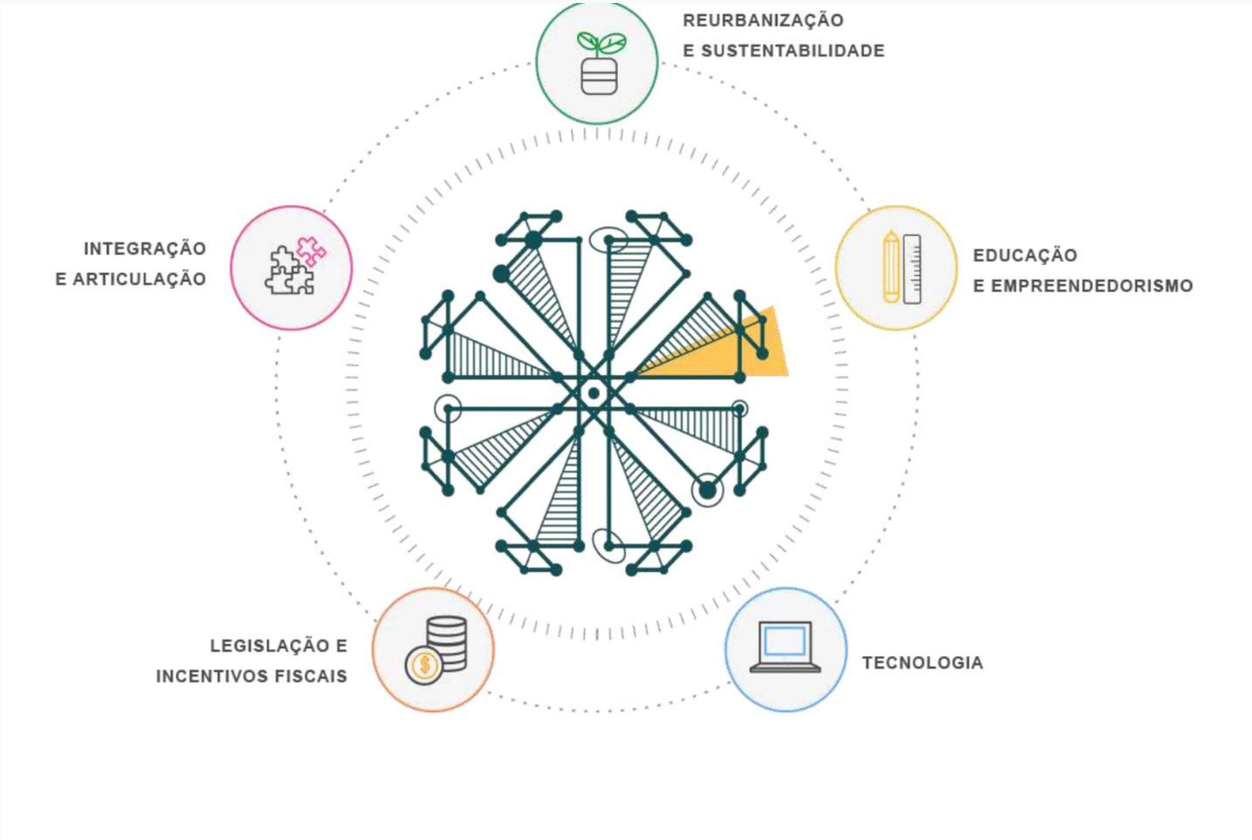
Farol do Saber

Neighborhood libraries and internet-access points broaden access to knowledge and digital services.

Vale do Pinhao

An innovation ecosystem that connects entrepreneurs, technology companies, research, and public policy.

Source in the original slide: valedopinhao.com.br/sobre/



Chef's Garden Program

- Works with 88 urban farmers from the Santa Rita IV Community Garden.
- Connects local production with sales to major city restaurants.

Ecocidadão Program

Improves the quality of life of waste pickers and strengthens the network for collecting and sorting reusable materials.

Entrepreneur Space

Provides guidance on formalizing a business, developing management skills, and accessing business advisory and consulting services.

Worktiba

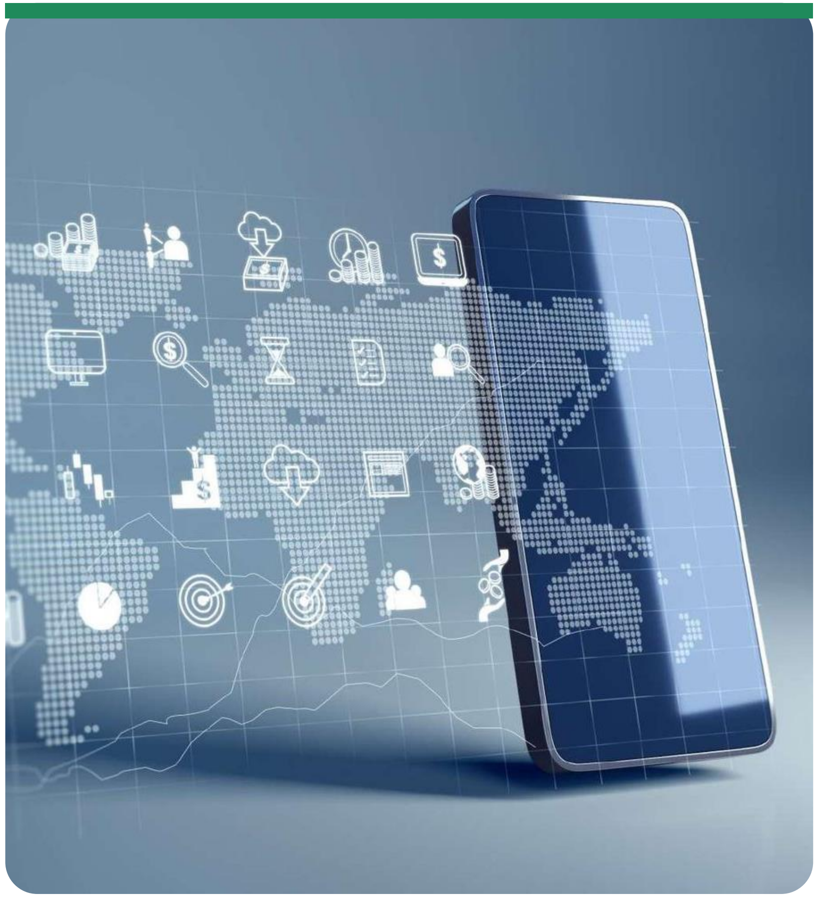
A shared workplace created to encourage innovative ideas, help them develop business potential, and generate income, employment, and new investment across Curitiba over the medium and long term.

Tecnoparque

A municipal program managed by the Curitiba Development Agency to:
- Support technology-based companies and science and technology institutions.
- Spread a culture of knowledge and innovation in strategic high-technology sectors.

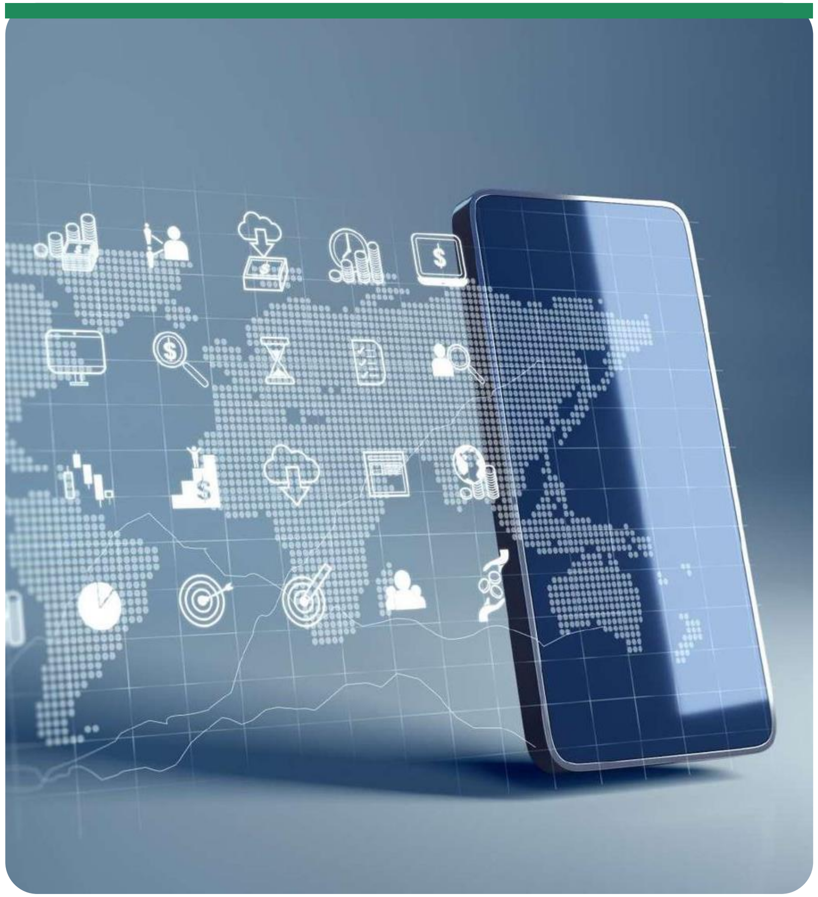
Curitiba App

- A simple, accessible search environment.
- Shows locations and information for municipal facilities, tourist attractions, and the city's events calendar.



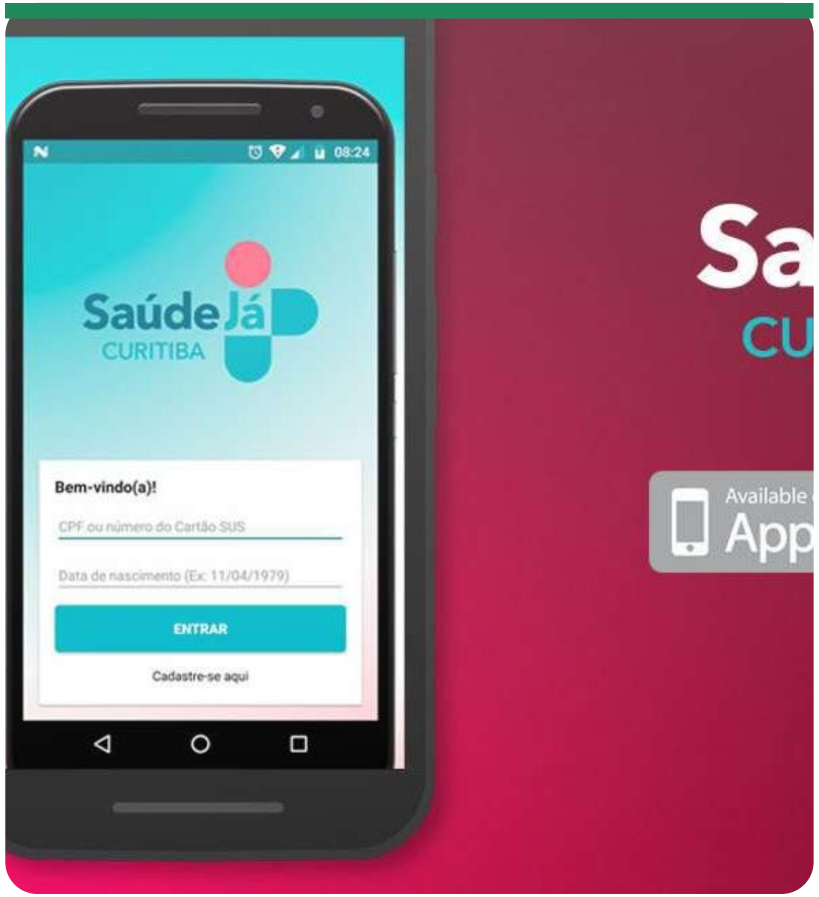
Curitiba 156 App

- A multichannel platform managed by the Municipal Government Secretariat.
- Speeds citizen interaction with municipal departments.
- Supports service evaluation, information search, and protocol tracking.



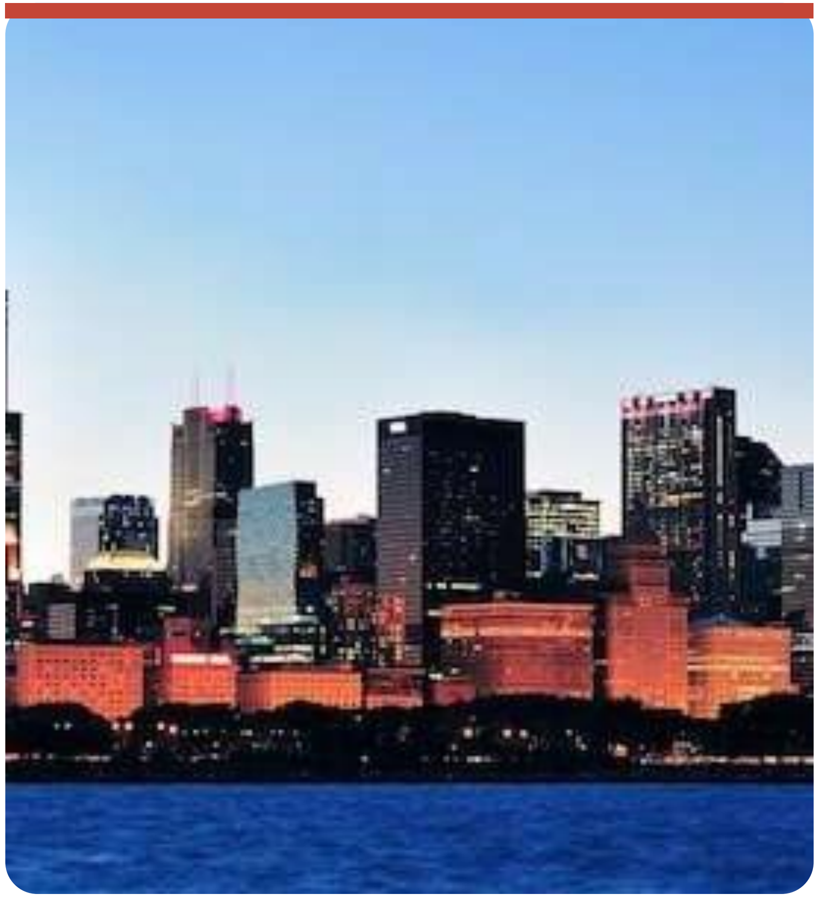
Saude Ja App

Allows residents to schedule their first appointment at a Municipal Health Unit - the entry point to Brazil's Unified Health System (SUS) in Curitiba.



Chicago at a glance

- Third-largest city in the United States.
- Population in the source material: 2.697 million.
- A major U.S. business center.
- Known for skyscrapers, museums, Impressionist art, and its history involving Al Capone.

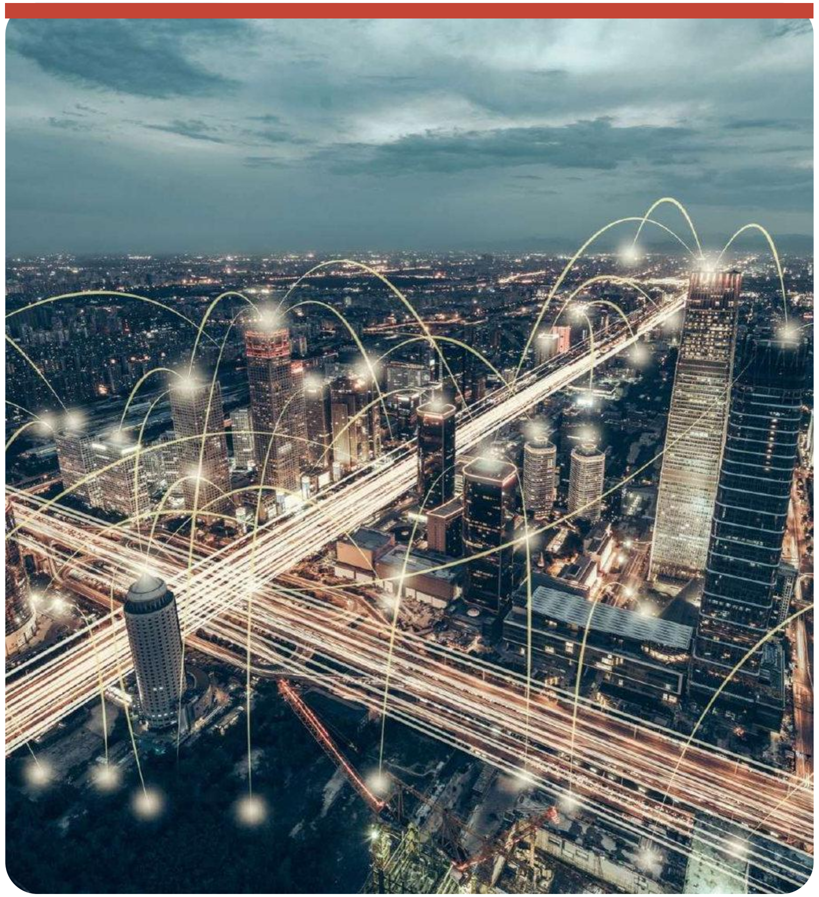


A leading U.S. smart city

Chicago is recognized as one of the United States' leading smart cities and has been at the forefront of applying innovative technologies to improve urban life.

Array of Things

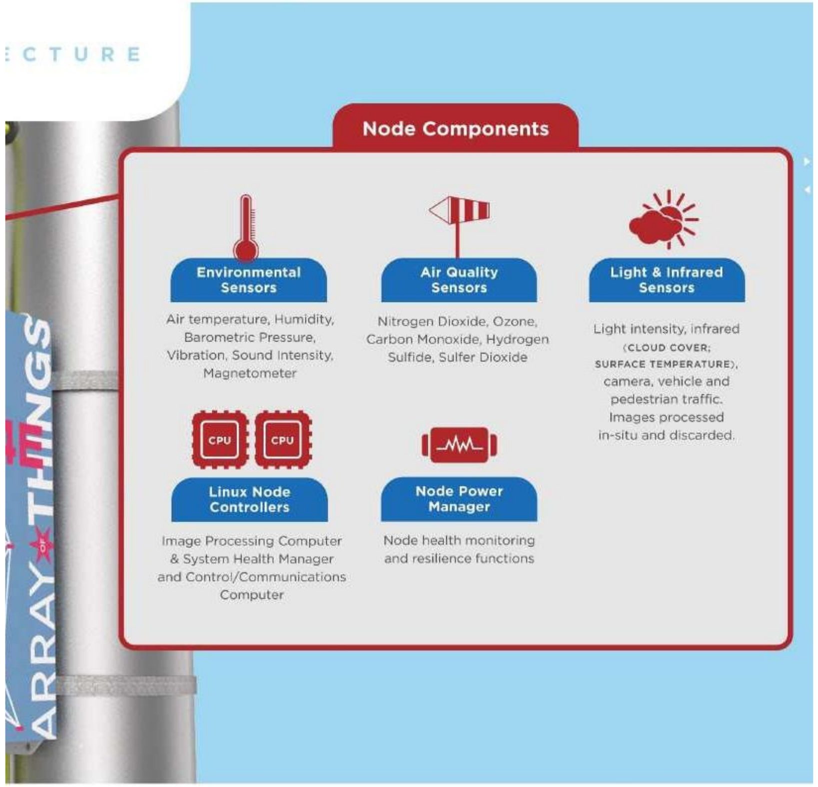
Array of Things (AoT) is an experimental urban-measurement project. A network of modular, interactive nodes collects real-time data about Chicago's environment, infrastructure, and activity.



Urban sensors measure the environment

Sensors measure temperature, barometric pressure, light, vibration, particulate matter, carbon monoxide, nitrogen dioxide, sulfur dioxide, ozone, and ambient noise.

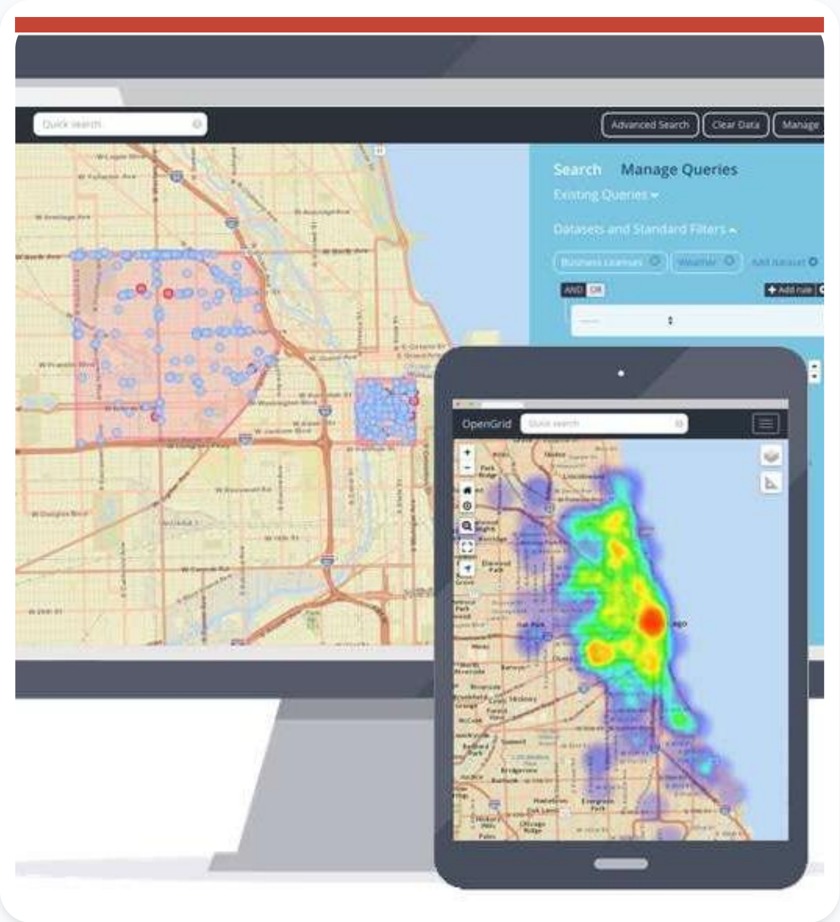
Source in the original slide: arrayofthings.github.io/



OpenGrid makes city data easier to explore

OpenGrid is a map-based application developed by Chicago's Department of Innovation and Technology. It helps residents visualize complex municipal data and engage with the city's changing patterns.

Source in the original slide: opengrid.io/



Smart street lighting at scale

- Described in the source material as the largest wireless smart street-lighting program in the United States.
- 280,000 public lights.
- Automatic alerts for power failures or burned-out lamps.

Chicago set long-term emissions targets

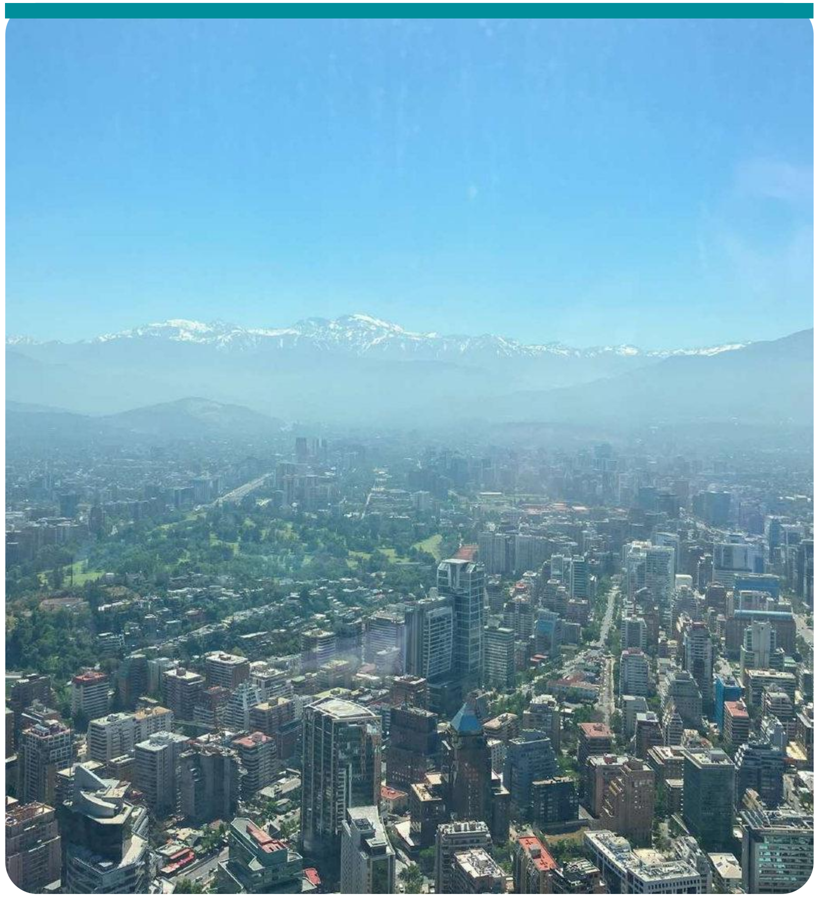
- In 2008, Chicago launched a plan targeting an 80% reduction in greenhouse-gas emissions.
- In 2022, the city committed to reducing carbon emissions by 62% by 2040.

Chicago's climate action plan: five pillars

- Expand access to utility savings and renewable energy, prioritizing families.
- Build circular economies that create jobs and reduce waste.
- Deliver a robust zero-emission mobility network.
- Advance equitable clean-energy development.
- Strengthen communities and protect health.

Santiago at a glance

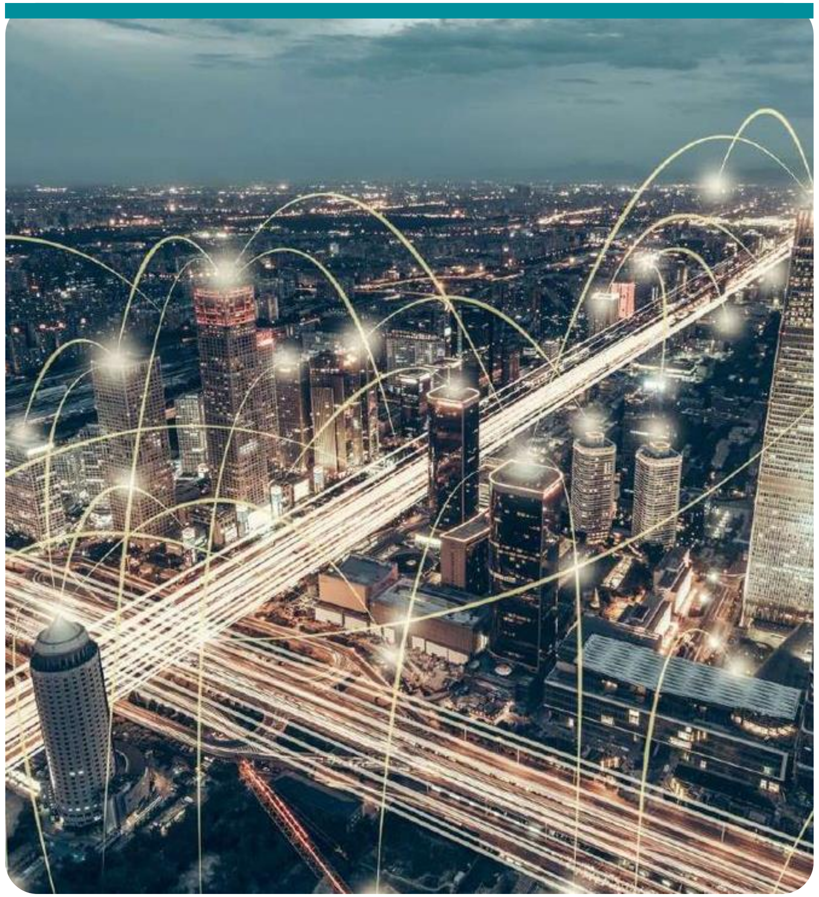
- Capital and largest city of Chile.
- Population in the source material: 5.614 million.
- Associated with poet Pablo Neruda.
- Surrounded by the Andes Mountains.



Chile's smart-city goals go beyond technology

- Transition fully to clean, renewable energy.
- Expand access to technology.
- Improve communication efficiency.

A regional smart-city reference



Santiago is described in the source material as Latin America's smartest city. Initiatives include:
- Environmental measures and surveillance cameras.
- Free Wi-Fi.
- Air-quality microsensors.
- A national-scale smart-city alliance.

Mobility and resilience initiatives

- BikeSantiago: the first public bicycle system.
- Red and Green Routes: metro-route management to reduce travel times.
- Reversible roads that change direction during specific periods.
- Development of a 'Resilient Santiago' strategy.

Connected services improve urban response

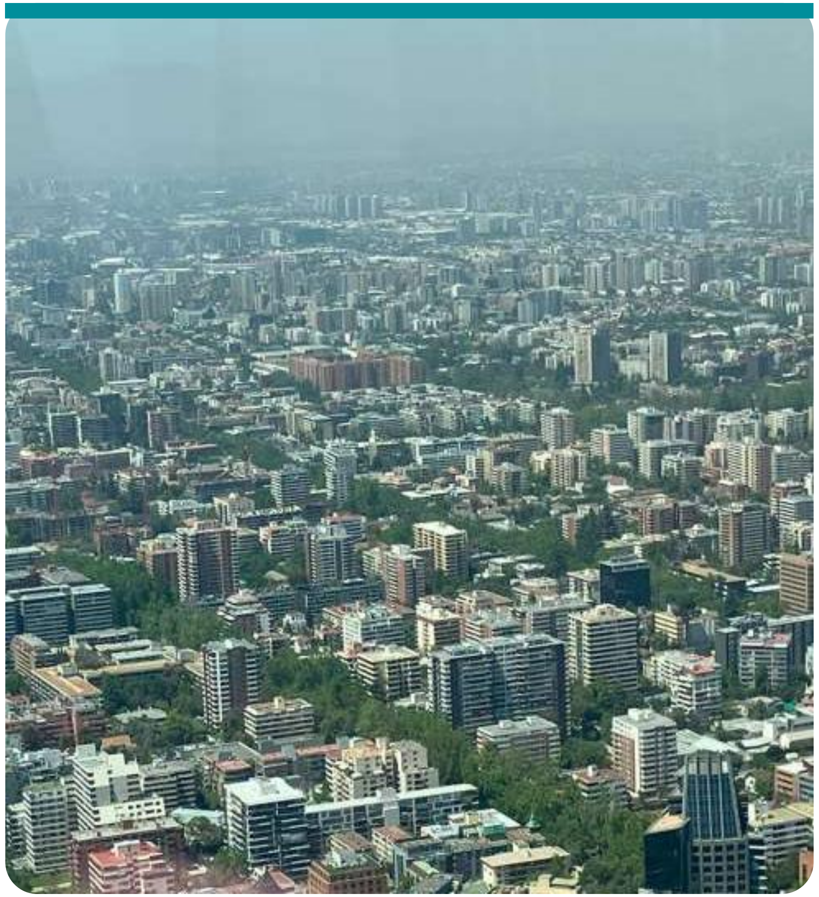
- Emergency coordination reduced waiting time by 30%.
- An air-quality app informs citizens about current conditions and vehicle restrictions.
- Corfo supports entrepreneurship and smart-city innovation.
- Telepeaje-TAG provides electronic toll collection based on identification technology.

SmartCity Santiago: Paseo Bandera

Paseo Bandera demonstrates how public-space design, culture, mobility, and technology can be combined in a central urban corridor.

SmartCity Santiago: air pollution

Air-quality monitoring supports public information, environmental management, and faster responses to pollution events.



Solar-powered air-quality microsensors

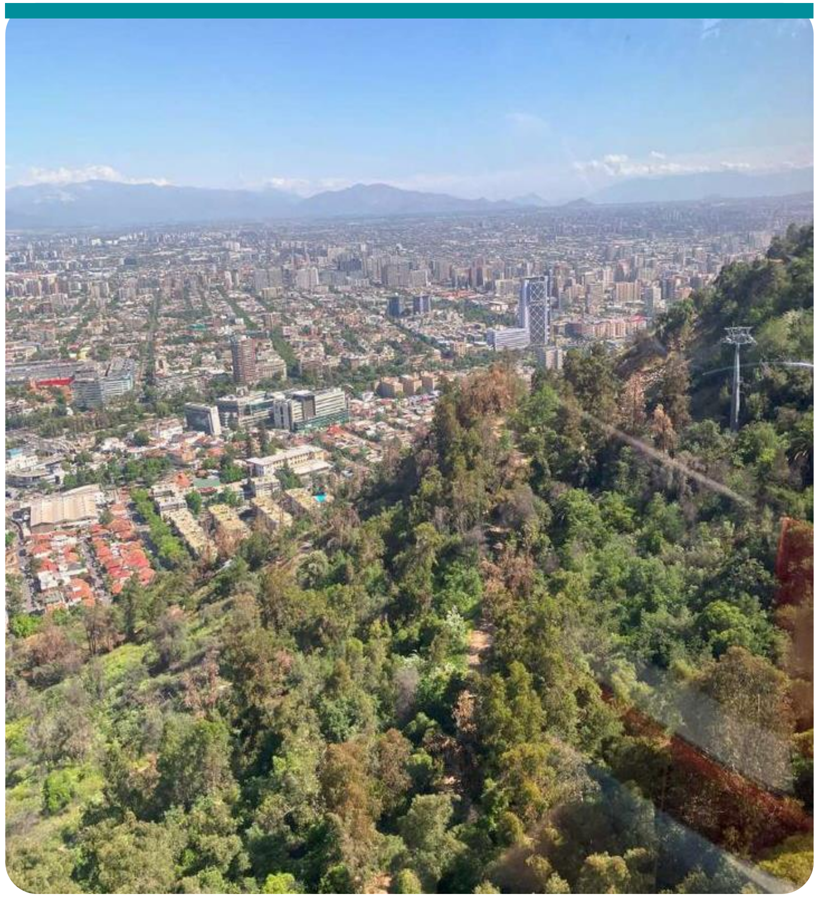
- Measure PM2.5, NOx, CO_2, VOCs, and other indicators.
- Send results in real time over cellular networks for analysis.
- Operate autonomously using solar panels.
- Offer low acquisition and maintenance costs with fast wireless installation.

Car-free streets on Sundays

Closing selected roads to cars on Sundays creates space for walking, cycling, recreation, and community life.

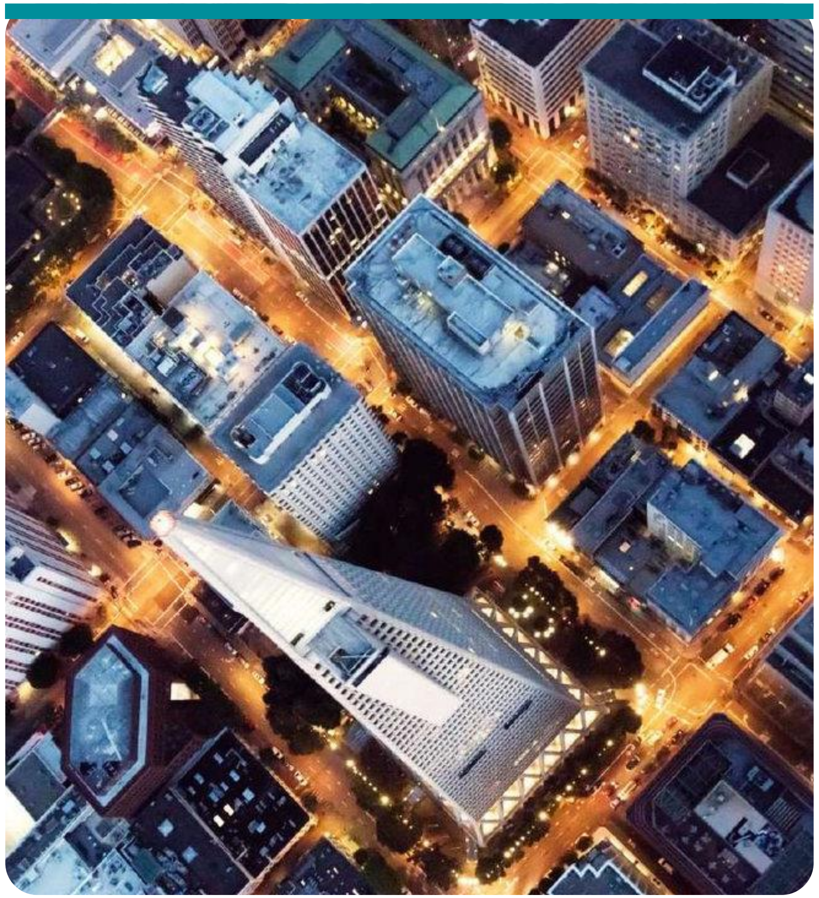
Green areas

Urban green spaces support recreation, biodiversity, thermal comfort, air quality, and climate resilience.



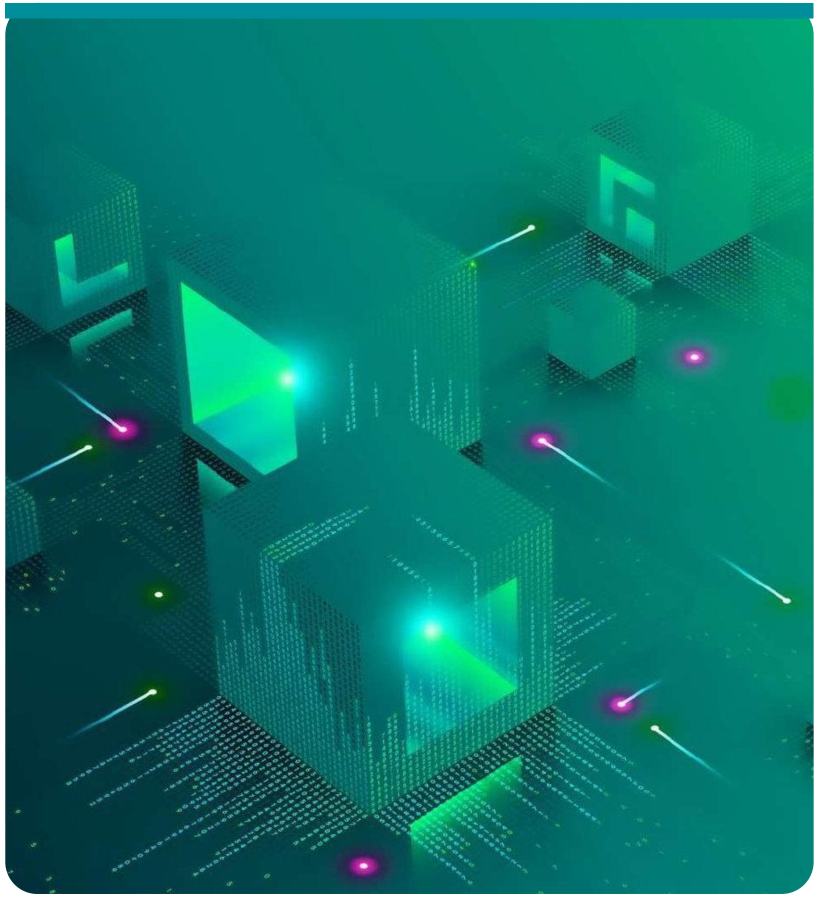
Ciudad Inteligente a Escala

- A smaller-scale environment that simulates a smart city.
- Developed by Claro Chile, the Pontifical Catholic University Innovation Center, and SE Santiago.
- Chile's first CIE was created at the San Joaquin Campus between 2022 and 2023.



One campus, many connected systems

- Transportation, lighting, security, agriculture, and forestry systems coexist on the UC San Joaquin Campus.
- The initiative combines 5G, data analytics, and artificial intelligence.



Connected safety systems

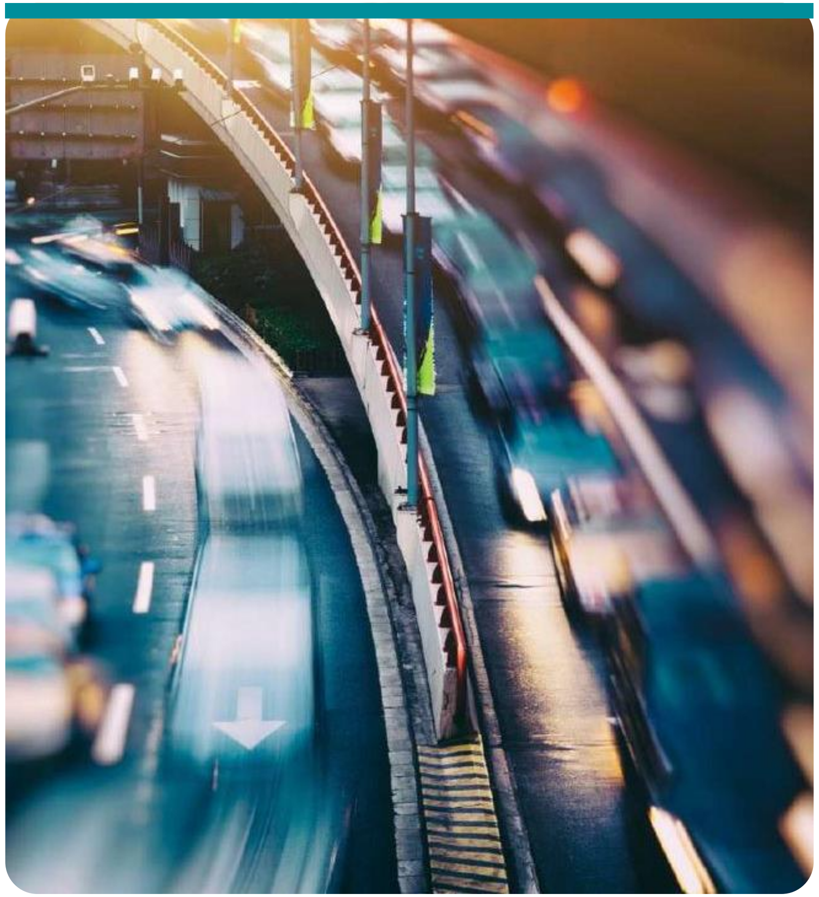
- Video analytics support physical security indoors and outdoors.
- QR codes enable rapid emergency response.
- Smart lighting operates together with security cameras.

Environmental and agricultural monitoring

- Multispectral cameras support environmental monitoring.
- Connected systems monitor beehives.
- Internet of Things sensors enable smart agriculture.

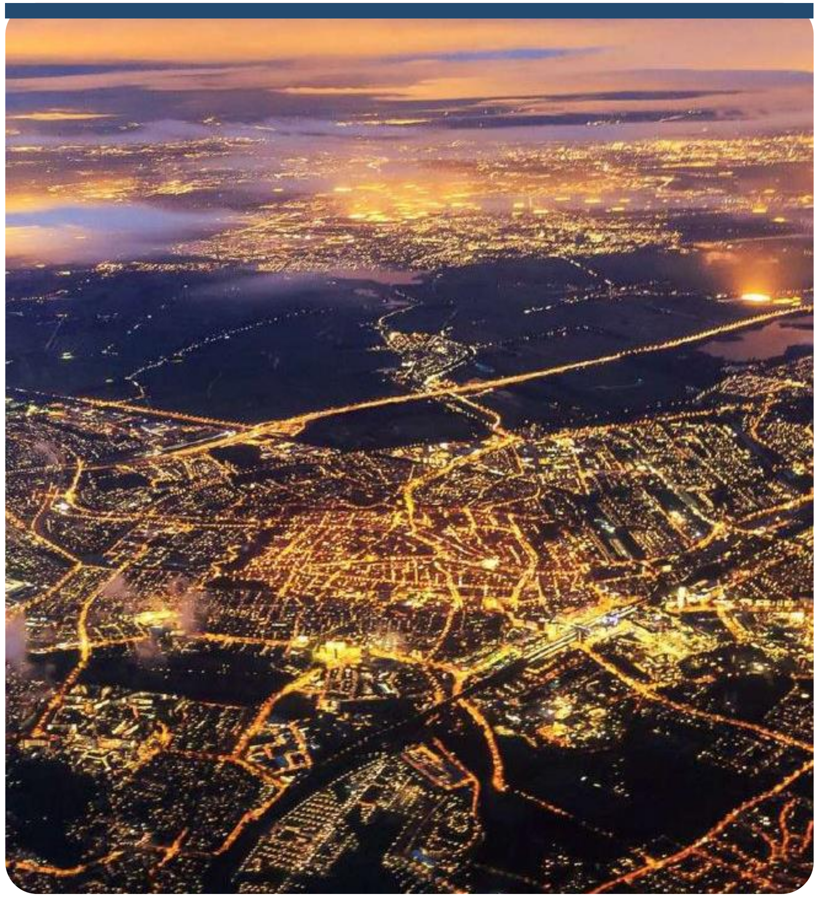
Mobility intelligence

- Vehicle traceability.
- A system for modeling flows of people and automobiles.



London at a glance

- Capital of England and the United Kingdom.
- A major global city and one of the world's most influential financial centers.
- Population in the source material: 8.982 million.
- Heathrow is described as the world's busiest airport.
- London has a thriving technology ecosystem.



Three-time Olympic host

London hosted the Summer Olympic Games in 1908, 1948, and 2012.

Intelligent traffic management

- A broad network of sensors and cameras monitors traffic in real time.
- Data optimizes traffic-signal patterns.
- Drivers receive immediate feedback on less-congested routes.
- Analysis identifies where infrastructure improvements are needed.

Policies and sensing manage congestion

- A daily congestion charge has applied to vehicles since 2003.
- High-polluting vehicles are restricted from designated zones.
- More than 1,500 cameras support enforcement, while CCTV sensors help analyze traffic flow.

Pedestrian-priority traffic signals

- Implemented at 18 intersections.
- The pedestrian signal remains green by default.
- It changes to red only when a sensor detects an approaching vehicle.

Smart parking

- Apps and sensors help drivers find available spaces quickly.
- This reduces circulating traffic and saves time.
- The source material reports 3,000 sensors installed in central-city parking spaces.

Environmental initiative: air-quality monitoring

London deploys sophisticated air-quality sensors across the city. The data identifies pollution hotspots, informs emissions policy, and enables alerts for residents with respiratory conditions.

London's long-term environmental ambitions

The source material highlights goals that include carbon neutrality by 2050, zero waste by 2050, and net-zero carbon by 2030. It also describes a target for more than half of London to be green by 2050.

Waste-reduction targets

- Reduce food and packaging waste by 50% per person by 2030.
- Reach a 50% recycling rate for waste collected by local authorities.
- Stop sending biodegradable and recyclable waste to landfill by 2026.
- Manage 100% of waste produced by 2026.

Buildings must move toward net-zero

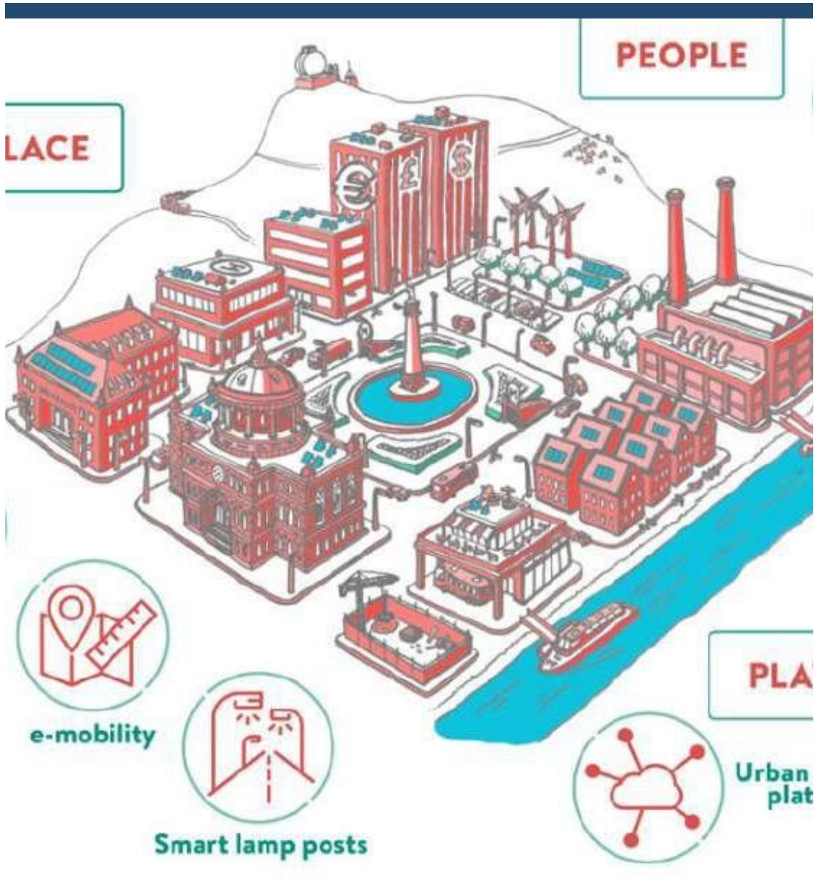
- New buildings should meet zero-carbon standards and operate at net-zero emissions by 2030.
- Existing buildings should be converted to net-zero emissions by 2050.

The Green Grid

London is developing a network of green spaces and corridors connecting parks, gardens, and other natural areas. The project aims to improve air quality, reduce the urban heat-island effect, and expand access to nature.

Sharing Cities initiative

- Focuses on sustainable transportation, energy efficiency, and citizen engagement.
- Includes pilots for electric-vehicle charging, smart street lighting, and community-energy initiatives.



Smart waste management

Sensors in waste containers monitor fill levels. This enables optimized collection routes, reduces fuel consumption, and helps prevent overflowing bins.

Public safety and services

Predictive analytics can help authorities identify potential crime hotspots, optimize police patrol routes, and improve overall public safety.

Smart street lighting

Street lights dim when no one is nearby to save energy. They can also carry sensors for traffic monitoring or crime detection.

Free public Wi-Fi

- London deployed more than 7,500 free Wi-Fi access points across the city.
- The initiative helps residents and visitors remain connected and supports digital access to services.

Open-data policy

- London makes valuable city data available to the public.
- Openness encourages innovation in transportation, health, and education.
- Developers can create new applications and services from shared data.

Digital health

- Combines NHS data and digital transformation with citywide initiatives.
- Discusses the safe use of patient data to improve healthcare and medical research.
- Digital platforms improve communication, access to medical information, and remote health services.

Shenzhen: from fishing village to metropolis

- Transformed into a major metropolis in about 30 years.
- The first Chinese city to electrify its entire bus and taxi fleet.
- Health records are available through a network of health centers.
- Uses systems to monitor traffic and people.

China's social-credit system

- The idea of a social-credit system emerged in 2014.
- The stated plan is to reward approved behavior and penalize misconduct.
- Millions of people classified as discredited were reportedly barred from buying train or plane tickets for violations such as using expired tickets or smoking on trains.

Identifying pedestrians who cross on red

The source slide presents a Chinese system designed to display pedestrians who do not respect red traffic signals.

Video referenced in the original slide: youtube.com/watch?v=sxUSk_rXEXQ

Dubai at a glance

- The largest city and namesake emirate in the United Arab Emirates.
- Population in the source material: 2.262 million.
- Known for a highly developed economy, large skyscrapers, and broad avenues.

Digital Dubai Office

More than 130 initiatives were launched with government and private-sector partners, including:
- Dubai Data Initiative.
- Dubai Blockchain Strategy.
- Happiness Agenda.
- Dubai AI Roadmap.

Dubai Data Initiative

- Manage data as a strategic asset.
- Build a citywide data platform for open data and value-added services.
- Increase the availability of city data.
- Improve data governance and compliance.
- Cultivate a strong data community.

Dubai Blockchain Strategy

- Enable simple, secure, and protected transactions.
- Create the world's first blockchain-powered government.
- Implement blockchain use cases across sectors such as finance, education, and real estate.
- Move government transactions toward blockchain technologies.

Happiness Agenda

- Position Dubai as the happiest city in the world.
- Discover people's needs and aspirations.
- Inspire positive change and awareness.
- Encourage self-reflection and estimate citywide impact through a Happiness Index.

Dubai AI Roadmap

- Establish an artificial-intelligence laboratory.
- Integrate AI into government services and city experiences.
- Improve citizens' overall quality of life.
- Increase happiness and maximize visitor satisfaction.

Dubai Pulse

Dubai Pulse is a next-generation digital operating system designed for individuals, businesses, the private sector, and government. It combines municipal services, urban-data orchestration, digital identity, cloud capabilities, and IoT resources.

Dubai Ethical AI Toolkit

- Provides practical support across the city's ecosystem.
- Helps industry, academia, and individuals understand how artificial-intelligence systems can be used responsibly.

From principles to self-assessment

- The toolkit combines principles, guidelines, and a self-assessment tool for developers.
- It seeks broad agreement and adoption of shared policies for the ethical use of AI.

Dubai Smart City Accelerator

- A creatively designed 5,000 m^2 space in Dubai Silicon Oasis.
- Helps startups expand and demonstrate products to companies and government partners.
- Offers guidance from more than 100 experts and office space in Dubai.
- Provides seed funding and access to a global investor and corporate network.

District 2020

- A smart city built on the Expo 2020 site.
- Sustainable, people-centered urbanism.
- Planned for 145,000 people.

A 15-minute district

- 45,000 square meters of green areas and parks.
- Five kilometers of walking and jogging paths.
- Ten kilometers of cycle lanes.
- Four kilometers for autonomous vehicles.
- Designed around the 15-minute-city concept.

Smarter mobility ecosystems

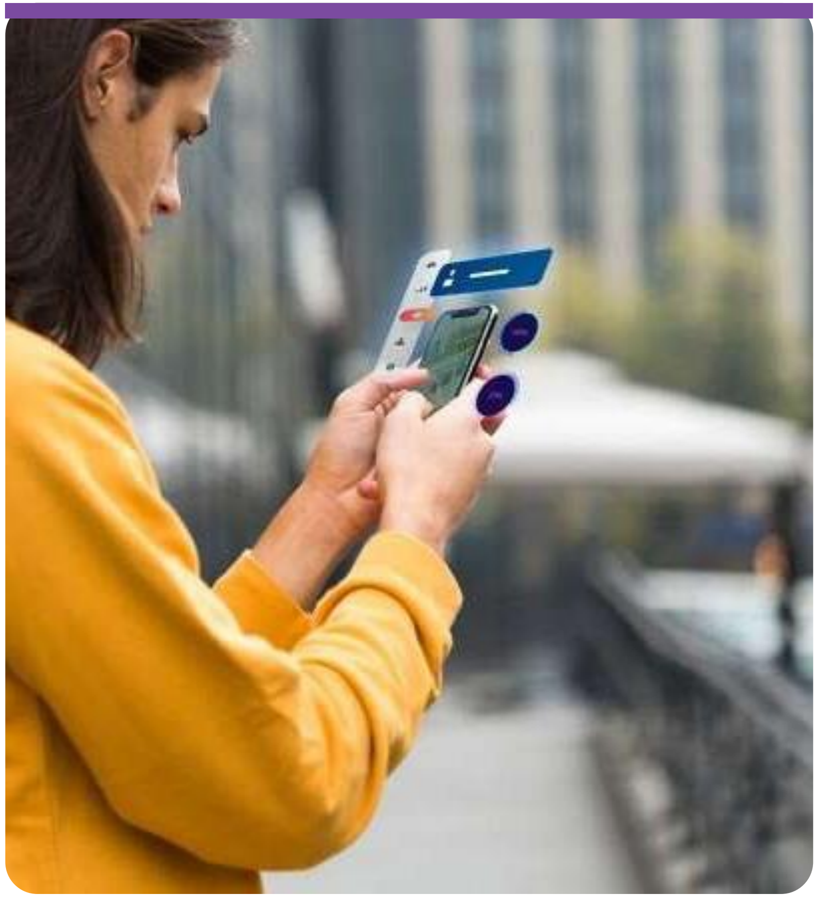
- Prioritize intelligent transportation initiatives.
- Expand the shared-vehicle market.
- Plan charging infrastructure to support electric-vehicle adoption.

Digital twins will strengthen connectivity

Digital twins can connect physical infrastructure, real-time data, simulation, and city operations.

Governments will expand digital services

State and local governments are expected to adopt more digital services, improving access, responsiveness, and integration.



Infrastructure must support climate-positive outcomes

- Build high-technology infrastructure that contributes positively to the climate.
- Integrate urban systems instead of operating them in isolation.

Data must produce meaningful outcomes

Digital technologies and urban data should be connected to relevant, measurable outcomes for people and communities.

Start with people, not technology

Effective smart-city strategies begin with human needs, public value, inclusion, and participation - then select the appropriate technologies.

The Line - Saudi Arabia

- NEOM introduced The Line as a human-centered futuristic-city concept.
- The source material describes a 34-square-kilometer area designed for approximately nine million residents.
- The project is associated with an estimated cost of US$500 billion.

The Line proposes five-minute access

- The concept aims to place essential facilities within a five-minute walk.
- A high-speed rail system is intended to connect the city with end-to-end travel of about 20 minutes.

The Line - reference sources

Original-slide sources:
- neom.com/en-us
- globalconstructionreview.com/why-saudi-arabias-the-line-is-not-a-revolution-in-urban-living/
- supercarblondie.com/saudi-arabia-the-line/

The Line - additional sources

Original-slide sources:
- youtube.com/watch?app=desktop&v=kYvOAXs5mOU
- middleeasteye.net/news/saudi-arabia-neom-line-satellite-images-progress-construction

Telosa - United States

- A proposed futuristic and utopian city developed by American entrepreneur Marc Lore and announced in September 2021.
- The project aims to house five million people by 2050, with an initial expectation of homes for 50,000.
- Planners considered a site in the western U.S. desert.

Telosa as a 15-minute city

- Workplaces, schools, and essential services are intended to be within 15 minutes of residents' homes.
- Fossil-fuel vehicles would be prohibited.
- Walking, scooters, bicycles, and autonomous electric vehicles would be prioritized.

Telosa - United States

The proposal combines urban planning, clean mobility, digital infrastructure, and a new model of city development.

Akon City - Senegal

- A project associated with Senegalese artist and entrepreneur Akon.
- The source material reports announcements beginning in 2018 and a further announcement in January 2022.
- The concept has been described as a 'real-life Wakanda' using blockchain and cryptocurrency.

Akon City's proposed infrastructure

- Condominiums, offices, parks, a university, an ocean resort, and a 5,000-bed hospital.
- Skyscrapers, shopping centers, a technology hub, music studios, 'Senewood,' and eco-tourism resorts.
- A smart, environmentally oriented city powered by renewable energy.

Akon City - reference

Original-slide source:
revistacasaejardim.globo.com/Casa-e-Jardim/Arquitetura/noticia/2021/01/rapper-americano-comeca-construcao-de-cidade-moderna-no-senegal.html